

Learning Adversarial Continuous MDPs with Bandit Feedback and Unknown Transitions

Abstract

We consider learning Markov Decision Processes (MDPs) with continuous state and action spaces, adversarial losses, bandit feedback, and unknown transitions. We present DiZOOM-PO, a policy optimization algorithm over a dynamically refined partition of the state-action space. It features three key ingredients: an adversarial zoom-in rule that refines the partition based on both cumulative and instantaneous confidences, dilated exploration bonuses adapted to the time-varying action sets, and a buffering mechanism that stabilizes importance-weighting estimators' bonuses on newly created regions. Together, we achieve $\tilde{\mathcal{O}}(K^{(z+1)/(z+2)})$ regret, where K is the number of episodes and z is an instance-dependent adversarial zooming dimension satisfying $z \leq d$ (the covering dimension). This improves over uniform discretization whenever $z < d$, matches the best known rate for stochastic continuous MDPs in the harder adversarial setting, and is tight in $\text{poly}(K)$ dependencies. We require neither simulators nor exploratory assumptions.

1 Introduction

Online reinforcement learning in finite-horizon episodic Markov Decision Processes (MDPs) with adversarial losses and bandit feedback admits several prominent algorithmic paradigms. The *occupancy-measure* paradigm (Zimin and Neu, 2013; Rosenberg and Mansour, 2019; Jin et al., 2020) runs online mirror descent over the convex polytope of all valid occupancy measures (the probability of visiting each state-action pair by a policy), achieving strong regret guarantees at the cost of solving a complicated convex optimization problem each round. This optimization problem has a size scaling with the cardinality of the state-action space, which is computationally expensive.

The *policy optimization* paradigm, on the other hand, instead updates on each state the probability of playing each action. It is computationally simple: in tabular (finite-state) MDPs it reduces to running separate Multi-Armed Bandit (MAB) algorithms at each state. Policy optimization originally traded simplicity for worse regret: Shani et al. (2020) achieve $\tilde{O}(K^{2/3})$ regret where K is the number of episodes, whereas $\tilde{O}(\sqrt{K})$ is achievable by occupancy-measure-based approaches. Luo et al. (2021) close this gap in the tabular setting by introducing *dilated exploration bonuses*: They design a modified Bellman equation that amplifies exploration at deeper layers and, consequently, cancels the variance due to adversarial importance-weighting estimators. Policy optimization has since been extended to adversarial MDPs with function approximation (Neu and Olkhovskaya, 2021; Dai et al., 2023; Liu et al., 2024), closing the regret gap with occupancy-measure methods in richer cases.

Neither paradigm, however, extends to *continuous* state and action spaces directly. Occupancy-measure methods rely on a polytope of occupancy measures that becomes measure-valued under continuous actions, while policy optimization must operate on a discretization whose resolution is difficult to choose. Uniform discretization incurs regret scaling with the *covering dimension* d (Kakade et al., 2003; Sinclair et al., 2019), which can be prohibitively large, while *adaptive discretization* circumvents this by refining only where near-optimal actions concentrate (Kleinberg, 2004; Kleinberg et al., 2019), achieving regret that scales with

an instance-dependent *zooming dimension* $z \leq d$. For stochastic continuous MDPs, [Cao and Krishnamurthy \(2020\)](#) obtained a zooming-dimension rate via a *buffering* mechanism, but their analysis relies on stationary rewards: an adversary can manipulate the statistics guiding discretization, forcing over-refinement of uninformative regions. In adversarial Lipschitz bandits, [Podimata and Slivkins \(2021\)](#) addressed this by augmenting the zoom-in rule with an *instantaneous* confidence condition that prevents splitting when the current action’s probability is low. Lifting this single-step technique to multi-layer adversarial MDPs remained open, primarily because the dynamic action set produced by zooming is inconsistent with the stable importance-weighting needed for policy optimization with bandit feedback. Therefore, our main question was:

Can we achieve provably-efficient learning in adversarial Lipschitz continuous MDPs under bandit feedback and unknown transitions?

We answer this in the affirmative, presenting DiZOOM-PO, an adversarial continuous MDP algorithm that performs policy optimization over a dynamically refined discretization of the state-action space. DiZOOM-PO achieves regret $\tilde{\mathcal{O}}(K^{(z+1)/(z+2)})$, where z is the adversarial zooming dimension of the instance. This rate is tight in K by the lower bound inherited from stochastic Lipschitz bandits ([Kleinberg et al., 2019](#)), strictly better than the uniform discretization technique whenever $z < d$, and matches the rate of [Cao and Krishnamurthy \(2020\)](#) for stochastic continuous MDPs in the strictly harder adversarial setting. Our algorithm requires neither transition simulators nor exploratory-policy assumptions.

Our algorithm combines dilated exploration bonuses for adversarial MDPs ([Luo et al., 2021](#)) with adversarial zooming for manipulation-robust adaptive discretization ([Podimata and Slivkins, 2021](#)). Naively combining them fails: adversarial zooming produces a time-varying action set, while dilated bonuses are defined over a stationary one. We therefore extend exponential weights, the Performance Difference Lemma ([Kakade and Langford, 2002](#)), and the dilated bonus analysis to dynamic state and action spaces. A further challenge is that newly appearing state-action pairs have intrinsically low sampling probability, which propagates across layers and breaks the dilated bonuses. Inspired by the buffering mechanism

for stochastic continuous MDPs (Cao and Krishnamurthy, 2020), we hold each new state-action pair in a “buffering set” until visited sufficiently often, and require that stepsize-scaled bonuses remain small on every active ball (Section 3.4). Together, these techniques achieve the near-optimal $\tilde{\mathcal{O}}(K^{(z+1)/(z+2)})$ regret in adversarial continuous MDPs with bandit feedback and unknown transitions.

Related work. For adversarial MDPs with infinite states, the most related line of research considers policy optimization in linear MDPs, where the simulator-free bandit-feedback rate has been progressively tightened from $\tilde{\mathcal{O}}(K^{14/15})$ (Luo et al., 2021) through a sequence of works (Dai et al., 2023; Sherman et al., 2023; Kong et al., 2024) to the current best $\tilde{\mathcal{O}}(K^{3/4})$ (Liu et al., 2024), with $\tilde{\mathcal{O}}(\sqrt{K})$ achievable under full information (Sherman et al., 2024). All such results strongly rely on the *linear* structure of Q -functions. On the continuous-RL side, Sinclair et al. (2019) initiated adaptive Q-learning for episodic metric MDPs at the worst-case covering-dimension rate, which Touati et al. (2020) and Cao and Krishnamurthy (2020) refined to the instance-dependent zooming dimension. The zooming idea itself dates back to stochastic Lipschitz bandits (Kleinberg, 2004; Bubeck et al., 2011; Kleinberg et al., 2019), with Podimata and Slivkins (2021) later resolving its adversarial counterpart; adversarial zooming for MDPs, to our knowledge, has not been developed.

A further detailed discussion of related work is deferred to Appendix A.1.

2 Preliminaries: Adversarial Continuous MDPs

Continuous metric MDPs. We consider learning in a finite-horizon episodic MDP $(\mathcal{S}, \mathcal{A}, H, \mathcal{P}, D)$ over $K > 0$ rounds. Here, $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, H is the horizon length, $\mathcal{P} = \{\mathcal{P}_h\}_{h=1}^H$ is the stochastic transition that is fixed across episodes, and D is a known metric over $\mathcal{S} \times \mathcal{A}$. We write $d_{\max} := \sup_{(x,a),(x',a') \in \mathcal{S} \times \mathcal{A}} D((x,a), (x',a'))$ for the diameter of the state-action space under D , which we assume is finite. For the ease of presentation, assume that the state space is layered as $\mathcal{S} = \bigcup_{h=1}^H \mathcal{S}_h$, such that the h^{th}

transition $\mathcal{P}_h$ is supported on $\mathcal{S}_{h+1}$. We further assume a singleton initial layer of $\mathcal{S}_1 = \{x_1\}$ (see, e.g., Jin et al., 2020). The state space $\mathcal{S}$ and action space $\mathcal{A}$ are continuous with infinite cardinality. The learner knows the metric D but not the transition $\mathcal{P}$.

The game is between a learner and an adversary. Before the game, an oblivious adversary commits in advance to a deterministic sequence of loss functions $\ell = \{\ell_k\}_{k \in [K]}$ with each $\ell_k: \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$.¹ The game proceeds over K episodes. Each episode $k \in [K]$ begins at x_1 and unfolds as follows:

1. The learner commits to a policy $\pi_k: \mathcal{S} \rightarrow \Delta(\mathcal{A})$ (the set of all probability distributions over $\mathcal{A}$).
2. For each layer $h = 1, \dots, H$, the learner observes state x_h^k , samples action $a_h^k \sim \pi_k(\cdot \mid x_h^k)$, suffers and observes loss $\ell_k(x_h^k, a_h^k)$, and transitions to the next state $x_{h+1}^k \sim \mathcal{P}_h(\cdot \mid x_h^k, a_h^k)$.

We consider the bandit feedback model, where only the losses suffered – namely $\{\ell_k(x_h^k, a_h^k)\}_{h \in [H]}$ – are revealed to the learner at the end of an episode. The full loss function ℓ_k remains hidden.

V-functions and Q-functions. Let Π denote the class of all policies $\pi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$. For any $\pi \in \Pi$ and any loss function $\ell: \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, define the V-function (the state-value function) and the Q-function (the state-action-value function) recursively as follows: $V_{H+1}^\pi(\cdot; \ell_k) \equiv 0$, and

$$Q_h^\pi(x, a; \ell_k) := \ell_k(x, a) + \mathbb{E}_{x' \sim \mathcal{P}_h(\cdot \mid x, a)} [V_{h+1}^\pi(x'; \ell_k)],$$

$$V_h^\pi(x; \ell_k) := \mathbb{E}_{a \sim \pi(\cdot \mid x)} [Q_h^\pi(x, a; \ell_k)], \forall h \in [H], x \in \mathcal{S}_h, a \in \mathcal{A}.$$

The learner’s objective is to minimize the expected cumulative *regret* against the best fixed policy in hindsight, namely a fixed $\pi^* \in \Pi$ that can adapt to the layer $h \in [H]$ but not

¹For a $n \in \mathbb{Z}_{\geq 0}$, $[n]$ denotes the set $\{1, 2, \dots, n\}$. We omit the layer subscript h in ℓ_k since $\mathcal{S}$ is already layered.

the episode $k \in [K]$:

$$\text{Reg}(K) := \sum_{k=1}^K V_1^{\pi_k}(x_1; \ell_k) - \inf_{\pi^* \in \Pi} \sum_{k=1}^K V_1^{\pi^*}(x_1; \ell_k).$$

For this hindsight optimum π^* , we write $Q_{k,h}^*(s, a) := Q_h^{\pi^*}(s, a; \ell_k)$ and $V_{k,h}^*(s) := V_h^{\pi^*}(s; \ell_k)$ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$, namely the Q-functions and the V-functions of π^* under the episode- k loss ℓ_k .

Lipschitz assumption. To enable adaptive discretization over $\mathcal{S} \times \mathcal{A}$, we require some continuity on the loss landscape and transition kernel, as without it, arbitrary spikes defeat any partitioning scheme. Following the standard Lipschitz framework in continuous RL (Kakade et al., 2003; Sinclair et al., 2019; Cao and Krishnamurthy, 2020), we make the following Lipschitzness assumption:

Assumption 1 (Lipschitz transitions and losses). *There exists some $L > 0$ such that for every $h \in [H]$ and every episode $k \in [K]$: (i) the loss ℓ_k is L -Lipschitz with respect to D ; and (ii) the transition P_h is L -Lipschitz in total variation:*

$$\|P_h(\cdot \mid x, a) - P_h(\cdot \mid x', a')\|_{\text{TV}} \leq L \cdot D((x, a), (x', a')), \quad \forall (x, a), (x', a') \in \mathcal{S}_h \times \mathcal{A}.$$

By the Bellman recursion, Assumption 1 implies that the Q-function $Q_{k,h}^{\pi}$ is $\mathcal{O}(LH)$ -Lipschitz for every policy $\pi \in \Pi$; see Remark 2 in the appendix for more details.

3 Dilated-Zoom Policy Optimization (DiZoom-PO) Algorithm

We present DiZoom-PO (DILATED-ZOOM POLICY OPTIMIZATION) in Algorithm 1. At a high level, the algorithm operates over a partition of the state-action space $\mathcal{S} \times \mathcal{A}$ – each element of which we call a “ball” – and dynamically refines this partition similarly to the

adversarial zooming technique of Podimata and Slivkins (2021). In an MDP, however, one not only needs to estimate the per-step losses of each ball, but also propagate value information across layers.

Two new challenges arise from this propagation. **(C1)** A newly created ball has not been visited in the past, so the importance-weighting estimator has a high variance and can mislead the policy update. **(C2)** The importance-weighting estimator’s variance scales inversely with the *learner’s* visit probability, while the regret decomposition reweights by the *comparator’s* visit probability; this q^*/q mismatch – after being amplified across the H layers – can be prohibitively large.

We resolve **(C1)** via a buffering-set technique inspired by Cao and Krishnamurthy (2020) (Lines 16 to 19 of Algorithm 1): newly split balls are placed in a “buffering set” to accumulate samples before promotion to the “active set,” as we introduce in Section 3.4. For **(C2)**, the dilated-bonus mechanism of Luo et al. (2021) addresses the q^*/q mismatch in tabular or linear MDPs by inflating exploration bonuses at deeper layers. But it only applies for stationary action sets. We extend it via the adversarial zooming rule of Podimata and Slivkins (2021) (Lines 10 and 13 of Algorithm 1), which keeps the amplified bonuses bounded under our adaptive discretization. We will detail this in Section 3.5.

3.1 Ball partitions and policies

To handle the continuous space, DiZOOM-PO maintains a hierarchical and dynamic partition of the state-action space $\mathcal{S} \times \mathcal{A}$. For each episode $k \in [K]$ and layer $h \in [H]$, we maintain a partition P_h^k of $\mathcal{S}_h \times \mathcal{A}$, whose elements we call *active balls*. Each ball $B \in P_h^k$ has a *representative center* $c(B) = (x_B, a_B) \in B$ and a radius $r(B)$ satisfying $D((x, a), c(B)) \leq r(B)$ for all $(x, a) \in B$.

At episode k and layer h , consider an arbitrary state $x \in \mathcal{S}_h$. Since P_h^k partitions $\mathcal{S}_h \times \mathcal{A}$, multiple active balls can contain x ; we denote this set by $\text{rel}_h^k(x)$ and call them the *relevant balls* of x . Each $B \in \text{rel}_h^k(x)$ covers a subset of actions $\text{act}_h^k(B; x) := \{a \in \mathcal{A} : (x, a) \in B\}$,

and the collection $\{\text{act}_h^k(B; x)\}_{B \in \text{rel}_h^k(x)}$ partitions $\mathcal{A}$. Our algorithm maintains a *ball policy* $\pi_k(\cdot | x) \in \Delta(\text{rel}_h^k(x))$; once a ball $B \in \text{rel}_h^k(x)$ is sampled from $\pi_k(\cdot | x)$, the agent plays the center action a_B .

In tabular MDPs, the occupancy measure $q_h^\pi(s, a) := \Pr(s_h = s, a_h = a | \pi)$ captures the probability of visiting each state-action pair at layer h (Jin et al., 2020). We extend this notion to our ball partition as follows. Let $q_{k,h}^*$ denote the law of x_h under the comparator policy π^* when the trajectory is generated by π^* and $\mathcal{P}$ starting from x_1 , viewed as a probability measure on $\mathcal{S}_h$. For a ball policy π , define its induced *ball occupancy* over balls as

$$q_{k,h}^{\star, \pi}(B) := \mathbb{P}\left((x_h, a_{B_h}) \in B \mid x_h \sim q_{k,h}^*, B_h \sim \pi(\cdot | x_h)\right),$$

that is, the probability of visiting ball B when playing π 's actions under π^* 's state occupancy. The learner's own ball occupancy $q_{k,h}^\pi(B)$ is defined identically but with x_h drawn from π 's state occupancy instead. We write $|P^k| := \sum_{h=1}^H |P_h^k|$ for the total number of active balls (summed across layers) at episode k , and $|P^K|$ for the final number of active balls.

When a ball is accurately estimated (Section 3.3), it is split into *child balls* to enable finer action selection – a “zoom-in” step. This creates a dynamic tree over balls. Since balls activate at different episodes, we write $\text{anc}_\tau(B)$ for the nearest tree ancestor of B that was active during episode τ .

3.2 Estimating transitions and Q-functions

Due to unknown transitions and bandit feedback, we estimate transitions and Q-functions online by adapting tabular MDP techniques (Jin et al., 2020; Luo et al., 2021) to our dynamic ball partition.

Confidence set of transitions. Let $n_h^k(B)$ be the number of visits to ball $B \in P_h^k$, or *one of its ancestors*, prior to episode k , and let $n_h^k(B, B')$ be the number of transitions from B to $B' \in P_{h+1}^k$. The empirical transition for episode k is $\hat{P}_h^k(B' | B) := \frac{n_h^k(B, B')}{n_h^k(B)}$ (with the

convention that $\frac{0}{0} = 0$).

As standard in tabular MDPs, we divide the K episodes into epochs $j = 1, 2, \dots$. A new epoch starts if any active ball has doubled its visit compared to the beginning of that epoch (see Line 11). At the beginning of each epoch j , a confidence set of transitions – namely $\mathcal{P}_j$ – is constructed:

$$\mathcal{P}_j := \{\hat{P} : |\hat{P}(B' | B) - \hat{P}_h^{k_j}(B' | B)| \leq \text{rad}_h^{k_j}(B, B'), \forall h, B, B'\}, \quad (1)$$

where k_j is the first episode of epoch j and $\text{rad}_h^{k_j}(B, B')$ is a Bernstein radius (see Appendix A.2).

Optimistic Q-functions. Let $L_{k,h} := \sum_{i \geq h} \ell_i^k$ be the total loss suffered in episode k from layer h onward. For each active ball $B \in P_h^k$, we use the upper-occupancy importance-weighting estimator:²

$$\hat{Q}_{k,h}(B) = \frac{L_{k,h} \mathbf{1}\{B = B_h^k\}}{\bar{q}_k(B) + \gamma_k}; \quad b_{k,h}(B) = \frac{3\gamma_k H + H(\bar{q}_k(B) - \underline{q}_k(B)) + (1 + 4 \log K)\beta_k}{\bar{q}_k(B) + \gamma_k}, \quad (2)$$

where $\bar{q}_k(B) := \max_{\hat{P} \in \mathcal{P}_j} q^{\hat{P}, \pi_k}(B)$ and $\underline{q}_k(B) := \min_{\hat{P} \in \mathcal{P}_j} q^{\hat{P}, \pi_k}(B)$ are the optimistic and pessimistic occupancies of a ball B within $\mathcal{P}_j$. The confidence radius $b_{k,h}(B)$ serves as a one-step exploration bonus that incorporates the bias of replacing the true occupancy $q_{k,h}^\pi$ with $\bar{q}_k$. We write $\gamma_{\min} := \inf_k \gamma_k = (2 + 4 \log K) \cdot 2H\eta_K$ for the smallest exploration rate across all episodes, which determines to the maximum estimation error and variance. More details are in Section 3.5.

3.3 Aggregate and instantaneous zoom-in rule

To dynamically refine the partition of $\mathcal{S} \times \mathcal{A}$, we adapt the adversarial zoom-in rule of Podimata and Slivkins (2021) to our multi-layer MDP problem. Specifically, for each active ball $B \in P_h^k$ at layer $h \in [H]$ of episode $k \in [K]$, we define the *aggregate confidence* and

²The notation $\mathbf{1}(E)$ is the indicator function for an event E .

instantaneous confidence as:

$$\text{conf}_h^{\text{tot},k}(B) := \frac{1}{\beta_k} + \sum_{\tau=1}^k \frac{\beta_\tau}{\pi_\tau(\text{anc}_\tau(B) \mid x_B)}, \quad \text{conf}_h^{\text{inst},k}(B) := \tilde{\beta}_k + \frac{\beta_k}{\pi_k(\text{anc}_k(B) \mid x_B)}, \quad (3)$$

where $\beta_k = \tilde{\beta}_k$ are defined in Line 3 of Alg 1. Intuitively, conf^{tot} accumulates the per-episode importance-weighting uncertainty β_τ/π_τ across the entire history of B , whereas $\text{conf}^{\text{inst}}$ captures only the episode- k contribution and is therefore sensitive to whether B is currently being sampled with nontrivial probability. A ball B is “zoomed in” at episode k if and only if both the aggregate and instantaneous conditions hold:

$$z_k(B) := \mathbf{1}\{\text{conf}_h^{\text{inst},k}(B) \leq e^{r(B)} - 1\} \cdot \mathbf{1}\{\text{conf}_h^{\text{tot},k}(B) \leq k \cdot r(B)\}. \quad (4)$$

The aggregate condition $\text{conf}_h^{\text{tot},k}(B) \leq k \cdot r(B)$ requires that the per-episode average of the cumulative estimation error has dropped to $r(B)$. Once this happens, the estimation error at B has saturated against the intrinsic spread of values within B (which can be as large as $L \cdot r(B)$ by Lipschitzness), so additional samples at the current resolution cannot sharpen our estimate further; the natural next step is to split B into finer children. The instantaneous condition $\text{conf}_h^{\text{inst},k}(B) \leq e^{r(B)} - 1$ (which is roughly $r(B)$ for small $r(B)$) requires that $\beta_k/\pi_k(B \mid x_B)$ is small, i.e., that the current sampling probability on B is sufficiently large. This rules out splitting based on rounds where π_k barely visits B , in which the adversary could have placed high losses without the algorithm noticing.

When both conditions are met, i.e., $z_k(B) = 1$, we replace B by its “children” $C(B)$, defined as an $r(B)/2$ -net of the domain of B . All the children $C \in C(B)$ also inherit B ’s cumulative bonus-corrected loss $\sum_{\tau < k} (\hat{Q}_{\tau,h}(B) - \mathcal{B}_{\tau,h}(B))$, which are used for policy optimization (Line 5). Equivalently, in the unnormalized exponential weights algorithm representation, the parent’s mass is split equally among its children. The children are placed into a “buffering set” F_h^{k+1} before replacing the parent ball B . We introduce this buffering scheme in Section 3.4.

Algorithm 1 DiZOOM-PO: Dilated-Zoom Policy Optimization

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1: Initialize: for each  $h$ ,  $P_h^1 \leftarrow$  a  $d_{\max}/H$ -net of  $\mathcal{S}_h \times \mathcal{A}$ ,  $F_h^1 \leftarrow \emptyset$ ; epoch index  $j \leftarrow 1$ ;
   confidence set  $\mathcal{P}_1$  as the set of all transition kernels over  $P_h^1$ . Fix  $\delta \in (0, 1)$ .
2: for each episode  $k = 1, 2, \dots, K$  do
3:   Set  $|P^k| = \sum_{h=1}^H |P_h^k|$ ,  $\eta_k = \min\{\frac{1}{24H^3}, (|P^k|HK)^{-1/2}\}$ ,  $\beta_k = \tilde{\beta}_k = \eta_k$ ,  $\gamma_k = (2 +$ 
      $4 \log K) \cdot 2H\eta_k$ .
4:   for each layer  $h = 1, 2, \dots, H$  do ▷ Play the “ball policy” defined in Section 3.1
5:     Observe  $x_h^k$ . Let  $\tilde{\pi}_k(B | x_h^k) \propto \exp\left(-\eta_k \sum_{\tau < k} (\hat{Q}_{\tau,h}(B) - \mathcal{B}_{\tau,h}(B))\right)$ ,  $\forall B \in \text{rel}_h^k(x_h^k)$ .
6:     Perform exploration w.p.  $\gamma_k$ :  $\pi_k(\cdot | x_h^k) \leftarrow (1 - \gamma_k)\tilde{\pi}_k(\cdot | x_h^k) + \gamma_k/|\text{rel}_h^k(x_h^k)|$ .
7:     Sample ball  $B_h^k \sim \pi_k(\cdot | x_h^k)$ . Play action  $a_h^k \leftarrow a_{B_h^k}$ . Observe loss  $\ell_h^k$ .
8:   for each layer  $h = H, \dots, 1$  do ▷ Update bonuses according to Section 3.5
9:     Construct Q-estimations  $\hat{Q}_{k,h}(B)$  and confidence radii  $b_{k,h}(B)$  according to Eq.
     (2).
10:    Calculate the dilated exploration bonus  $\mathcal{B}_{k,h}(B)$  from  $b_{k,h}(B)$  according to Eq. (5).
11:    Update visit counters and the transition confidence set according to Section 3.2; if
     any active ball has doubled its visits since epoch  $j$  began, increment  $j$  and refresh  $\mathcal{P}_j$ 
     as in Eq. (1).
12:   for each active ball  $B \in \bigcup_{h \in [H]} P_h^k$  do ▷ Zoom-in balls according to Section 3.3
13:     if  $z_k(B) = 1$  (defined in Eq. (4)) then
14:       Create “children balls”  $C(B)$  as an  $r(B)/2$ -net of  $\text{act}_h^k(B; x_B)$ .
15:       Each child  $C \in C(B)$  inherits  $B$ ’s cumulative bonus-corrected loss (Section 3.3).
16:       Place children in “buffering set”:  $F_h^{k+1} \leftarrow F_h^k \cup C(B)$ .
17:   for each buffered ball  $B' \in \bigcup_{h \in [H]} F_h^k$  do ▷ Activate balls according to Section 3.4
18:     if all children of  $\text{parent}(B')$  satisfy  $n_h^k(\cdot) \geq N_{\min}(\cdot)$  and  $\eta_k \mathcal{B}_{k,h}(\cdot) \leq 1/(2H)$  then
19:       Promote the entire sibling group to the active set, and remove the parent:
        $P_h^{k+1} \leftarrow (P_h^{k+1} \setminus \{\text{parent}(B')\}) \cup C(\text{parent}(B'))$ ,  $F_h^{k+1} \leftarrow F_h^{k+1} \setminus C(\text{parent}(B'))$ .

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3.4 Buffering set: stabilizing estimators on newly created balls

A child ball just produced by the zoom-in rule has, by construction, never been played on its own: its cumulative bonus-corrected loss $\sum_{\tau < k} (\hat{Q}_{\tau,h}(B) - \mathcal{B}_{\tau,h}(B))$ is inherited entirely from its parent. Including such a ball directly in policy optimization is dangerous: the importance-weighting estimator $\hat{Q}_{k,h}(B)$ on a freshly created ball has a very small denominator $\bar{q}_k(B) + \gamma_k$ (recall Eq. (2)), so a single loss can move π_k by an amount much larger than the bonus. The dilated-bonus argument, which requires an upper bound on the per-ball bonus (formalized soon in Lemma 2), then fails on low-mass balls.

To remedy this, we maintain a buffering set F_h^k for newly created balls. We require two

conditions before promoting a buffered $B' \in F_h^k$ to the active set P_h^k : (i) it must accumulate $N_{\min}(B') \approx r(B')^{-2}$ visits, the same condition that [Cao and Krishnamurthy \(2020\)](#) used in stochastic continuous cases; and (ii) the dilated bonus on B' , defined in the next section, must satisfy $\eta_k \mathcal{B}_{k,h}(B') \leq 1/(2H)$. Condition (i) controls the variance of importance-weighting on B' . Condition (ii) ensures that $\eta_k |\hat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B)| \leq 1$ holds for the loss vector fed into the exponential weights update at B' (see [Section 3.5](#)), which is needed for the standard exponential weights regret bound.

Sibling-group promotion. To maintain the partition invariant – at every layer h , the active set P_h^k partitions $\mathcal{S}_h \times \mathcal{A}$ – we promote children only as a complete sibling group. Concretely, once all children $C \in C(\text{parent}(B'))$ jointly satisfy the two promotion conditions above, we replace the parent by the entire sibling group atomically ([Line 19](#)); promoted-but-waiting children remain in the buffering set. This ensures that the parent and any of its (already promoted) children never coexist in P_h^k , which would otherwise violate the partition property. An equivalent alternative is to keep children buffered individually but always have π_k act on the finest active ancestor at each visited point; we adopt the sibling-group rule for cleaner accounting.

Note that $\mathcal{B}_{k,h}(B')$ is well-defined even for buffered B' . In particular, the dilated-bonus recursion evaluates backward from layer H to h , and at each layer the active set P_h^k is a partition (the sibling-group promotion rule above ensures partition coverage at every step). The bottom-up promotion order $h = H, \dots, 1$ guarantees that P_{h+1}^{k+1} is finalized before $\mathcal{B}_{k,h}(B')$ is evaluated.

When buffered, $B' \in F_h^k$ is invisible: π_k is only defined over the active set P_h^k , and visits to B' are accumulated only through visits to $\text{anc}_k(B')$ (tracked by the counter update in [Section 3.2](#)). Once the sibling group is jointly ready, all members join P_h^k and participate in the exponential weights update.

3.5 Exponential weights and dilated bonuses

At each visited state, the algorithm runs exponential weights locally over the relevant playing balls, with $\tilde{\pi}_k(B \mid x_h^k) \propto \exp(-\eta_k \sum_{\tau < k} (\hat{Q}_{\tau,h}(B) - \mathcal{B}_{\tau,h}(B)))$ as in Line 5. The standard analysis of policy optimization via exponential weights would, in the bandit-feedback continuous setting, leave a single-ball error inversely proportional to $q_k(B)$, which is the occupancy measure of B induced by our policy π_k . Summing this error across layers, it will be reweighed by the comparator’s occupancy measure $q^*(B)$, producing a ratio $q^*(B)/q_k(B)$ – which the adversary can amplify across the H layers (the deeper a state is, the more the mismatch compounds). The standard exploration techniques (explicit or implicit) only address single-step errors, but cannot cancel such cross-layer amplifications.

The dilated exploration bonuses technique of [Luo et al. \(2021\)](#) addresses this cross-layer amplification by increasing the bonus when propagating across layers, so that the layer- h bonus is much larger than the layer- $(h+1)$ residual error. We define a ball-level dilated bonus recursively:

$$\mathcal{B}_{k,h}(B) \geq b_{k,h}(B) + \left(1 + \frac{1}{H}\right) \max_{\hat{P} \in \mathcal{P}_j} \sum_{B' \in P_{h+1}^k} \hat{P}(B' \mid B) \sum_{C \in \text{rel}_{h+1}^k(x_{B'})} \pi_k(C \mid x_{B'}) \mathcal{B}_{k,h+1}(C), \quad (5)$$

where $b_{k,h}(B)$ is the one-step exploration bonus in Eq. (2) and $\mathcal{B}_{k,H+1} := 0$. The factor $1+1/H$ inflates the recursion just enough that, when the bonus is subtracted from the importance-weighting-style loss inside the exponential weights update, the deeper-layer residual is always cancelled by the shallower-layer bonus. The policy π_k is then derived over the relevant playing balls using exponential weights with the bonus-corrected loss estimator $\hat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B)$. As we will show in Lemma 2 (formalized in Appendix A.7), this construction reduces the total H -layer regret to a single layer-1 residual $\sum_k \hat{V}_1^{\pi_k}(x_1; b_k)$, which is bounded by the sum of one-step bonuses alone.

3.6 Main guarantee for DiZoom-PO

By combining these algorithmic components, we obtain the following guarantee for DiZoom-PO:

Theorem 1 (Main Guarantee for DiZoom-PO). *Under the Lipschitz Assumption 1 and using Algorithm 1 with the confidence-based zoom-in rule (4), with probability at least $1 - \delta$, we have*

$$\text{Reg}(K) = \tilde{\mathcal{O}}_K \left(K^{\frac{z+1}{z+2}} \right). \quad (6)$$

In Theorem 1, z is the adversarial zooming dimension of $\mathcal{S} \times \mathcal{A}$, formally defined in Appendix A.2. It never exceeds the covering dimension of $\mathcal{S} \times \mathcal{A}$, namely d . The $\tilde{\mathcal{O}}_K$ notation hides $\text{polylog}(K)$ factors and the polynomial dependencies on H and $\log(1/\delta)$. The full regret bound, including the dependence on L , H , $\log(1/\delta)$, and the geometric constants of the metric, is presented in Appendix A.2. The full proof is in Appendix A.4, and we sketch the main steps in the next section.

Because $z \leq d$, Theorem 1 is never worse than the $\tilde{\mathcal{O}}(K^{(d+1)/(d+2)})$ rate attained by uniform discretization (actually strictly better whenever $z < d$). Eq. (6) also matches the best bound for stochastic continuous MDPs (Cao and Krishnamurthy, 2020) – but in our strictly harder adversarial setting. It is tight in K (up to logarithmic factors) by the $\Omega(T^{(z+1)/(z+2)})$ lower bound for stochastic Lipschitz bandits (Kleinberg et al., 2019). As a special case, setting $d = z = 0$ recovers $\tilde{\mathcal{O}}(H^{3/2} \sqrt{|\mathcal{S}||\mathcal{A}|K})$ regret for tabular adversarial MDPs, matching the rate of Jin et al. (2020) up to $\text{poly}(H)$ factors.

4 Analysis Sketch of DiZoom-PO

The regret is evaluated on a continuous MDP, but our DiZoom-PO only operates on discrete sets of balls. The analysis proceeds in three steps: (i) a *three-term decomposition* of the ball-level regret via the Performance Difference Lemma into BIAS-1, BIAS-2, and REG-

TERM, each individually bounded; (ii) a *layer-wise cancellation* via the dilated bonuses that collapses the cross-layer accumulation of importance-weighting variance into a single sum of one-step bonuses (Lemma 2); and (iii) a *continuous-to-discrete reduction* via Lipschitzness that combines the ball-level bound with the discretization error to achieve the final regret guarantee (Lemma 3).

4.1 Ball-level three-term decomposition

The idea behind any policy optimization algorithm is the Performance Difference Lemma (Kakade et al., 2003), which relates the global regret to a per-state local optimization error. We extend this device to ball-level: restricting the comparison to the (finite, time-varying) set of active balls converts the infinite-dimensional problem into a sequence of finite bandit problems indexed by layer.

Two new challenges, however, complicate the standard analysis. (i) Bandit feedback in a continuous space requires importance-weighting estimators with potentially unbounded variance; and (ii) adaptive discretization changes the action set across rounds, breaking standard fixed-action exponential weights bounds. For ball-policy π , define the representative value $Q_{k,h}^\pi(B) := Q_{k,h}^\pi(c(B))$. By the standard performance-difference lemma applied at ball level, the ball-level regret $\sum_{k,h,B} (q_{k,h}^{\star,\pi_k}(B) - q_{k,h}^{\star,\pi^\star}(B)) Q_{k,h}^{\pi_k}(B)$ can be rewritten with an inserted $\pm \mathcal{B}_{k,h}(B)$ (which telescopes; see Lemma 2) as

$$\sum_{k=1}^K \sum_{h=1}^H \sum_{B \in P_h^k} q_{k,h}^{\star,\pi_k}(B) (\pi_k(B | x_B) - \pi^\star(B | x_B)) (Q_{k,h}^{\pi_k}(B) - \mathcal{B}_{k,h}(B)).$$

Adding and subtracting the importance-weighted estimator $\widehat{Q}_{k,h}(B)$ inside the parentheses then decomposes this expression into three terms:

$$\underbrace{\sum_{k,h,B} q_{k,h}^{\star,\pi_k}(B) (Q_{k,h}^{\pi_k}(B) - \widehat{Q}_{k,h}(B))}_{\text{BIAS-1: under-estimation error}} + \underbrace{\sum_{k,h,B} q_{k,h}^{\star,\pi^\star}(B) (\widehat{Q}_{k,h}(B) - Q_{k,h}^{\pi_k}(B))}_{\text{BIAS-2: over-estimation error}}$$

$$+ \underbrace{\sum_{k,h,B} (q_{k,h}^{\star,\pi_k}(B) - q_{k,h}^{\star,\pi^*}(B)) (\hat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B))}_{\text{REG-TERM: exponential weights regret}}. \quad (7)$$

We bound BIAS-1 (Lemma 11), BIAS-2 (Lemma 12), and REG-TERM (Lemma 10) in turn.

Bounding Bias-1 and Bias-2. BIAS-1 and BIAS-2 can be controlled in now-standard ways: the optimism used in $\hat{Q}_{k,h}$ ensures that BIAS-2 is non-positive in expectation, and BIAS-1 is well-controlled because the one-step bonus $b_{k,h}(B)$ in Eq. (2) always cancels the estimation error. In particular, the bonus $b_{k,h}$ is designed to incorporate the bias from replacing the true occupancy $q_k(B)$ by the optimistic occupancy $\bar{q}_k(B)$ in the importance-weighting denominator: this is what the $H(\bar{q}_k - \underline{q}_k)/(\bar{q}_k + \gamma_k)$ term in Eq. (2) buys us. See Lemmas 11 and 12 for the formal arguments.

Bounding Reg-Term. REG-TERM suffers from the variance of the importance-weighting estimator – scaling inversely in the learner’s visit probability $q_k(B)$ – but reweighed by the comparator’s visit probability $q^*(B)$. In addition to the dilated bonuses detailed soon, two modifications to standard exponential-weight analysis are needed: (a) we use a varying action-set variant of exponential weights, which incorporates the cost of newly created child balls into an extra ancestor-tree term in the regret bound; and (b) the buffering condition $\eta_k \mathcal{B}_{k,h}(B) \leq 1/(2H)$ on every active ball (recall Section 3.5) controls their otherwise unstable variance, thus bounding the maximum bonus across active balls.

4.2 Layer-wise cancellation via dilated bonuses

The mismatch between the comparator occupancy $q^*(B)$ and the learner occupancies $q_k(B)$ in REG-TERM would, if without further intervention, amplify across the H layers and become exponentially large. To resolve this issue, expanding the performance difference one layer at a time gives

$$V_{k,h}^{\pi_k} - V_{k,h}^{\pi^*} = \mathbb{E}_{a \sim \pi_k} [Q_{k,h}^{\pi_k}(a)] - \mathbb{E}_{a \sim \pi^*} [Q_{k,h}^{\pi^*}(a)]$$

$$\approx (\text{layer-}h \text{ local error}) + \underbrace{\mathbb{E}_{a \sim \pi_k} [V_{k,h+1}^{\pi_k} - V_{k,h+1}^{\pi^*}]}_{\text{propagates to layer } h+1},$$

where the second term recurses under the learner's policy π_k . The dilated bonus $\mathcal{B}_{k,h}$ in Eq. (5) mirrors this exact recursing term, with an extra $(1 + 1/H)$ factor introduced to ensure the layer- h bonus is slightly larger than the propagated residual from layer $h + 1$. The error then telescopes backward from layer H to layer 1, leaving only the residual term at the initial state. This is consequently bounded by the sum of one-step bonuses, $\widehat{V}_1(x_1; b_k)$. We collect these in the following lemma.

Lemma 2 (Ball-level dilated-bonus cancellation). *Combining the bounds on BIAS-1, BIAS-2, REG-TERM with the dilated-bonus construction (5) gives*

$$\sum_{k=1}^K \sum_{h=1}^H \sum_{B \in P_h^k} \left(q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B) \right) Q_{k,h}^{\pi_k}(B) \leq \widetilde{\mathcal{O}}(H/\eta_K) + 4 \sum_{k=1}^K \widehat{V}_1^{\pi_k}(x_1; b_k).$$

Lemma 2 therefore reduces the entire ball-level regret to $\widetilde{\mathcal{O}}(H/\eta_K)$ plus a sum of optimistic one-step bonus values $\widehat{V}_1^{\pi_k}(x_1; b_k)$. Its formal proof is deferred to Appendix A.7. It only remains to connect this ball-level bound back to the original regret, and also control the sum of one-step bonuses.

4.3 Continuous-to-discrete reduction and final bound

The regret is defined on the continuous MDP, but the algorithm only acts on the active partition. According to Assumption 1, each ball B may contribute an additional discretization error up to $L \cdot r(B)$, weighted by either $q_{k,h}^{\star, \pi_k}$ or $q_{k,h}^{\star, \pi^*}$ along the trajectories. Combining with Lemma 2 gives:

Lemma 3 (Continuous-to-discrete reduction). *Under Assumption 1,*

$$\text{Reg}(K) \leq \widetilde{\mathcal{O}}(H/\eta_K) + 4 \sum_{k=1}^K \widehat{V}_1^{\pi_k}(x_1; b_k) + L \sum_{k,h,B} (q_{k,h}^{\star, \pi_k}(B) + q_{k,h}^{\star, \pi^*}(B)) r(B),$$

The proof of Lemma 3 is in Appendix A.6. To prove the main theorem, we then need to control the bonus sum and the discretization error. After the bonus $b_{k,h}$ incorporates the estimation bias from BIAS-1, each ball contributes at most γ_k to the per-layer sum $\sum_B q_k(B)/(\bar{q}_k(B) + \gamma_k)$, so the per-layer bonus is bounded by $\gamma_k H |P_h^k| = \tilde{\mathcal{O}}(\eta_k H^2 |P_h^k|)$. Summing over layers and episodes gives

$$\sum_{k=1}^K \hat{V}_1^{\pi_k}(x_1; b_k) \leq \tilde{\mathcal{O}}(H^{3/2} \sqrt{|P^K|K}). \quad (8)$$

For the discretization error, we introduce an undertermined threshold $\varepsilon_S > 0$ and split by scale: “refined balls” with $r(B) \leq \varepsilon_S$ contribute $\mathcal{O}(LH^3 K \varepsilon_S)$, while “coarse balls” are visited rarely. In Appendix A.5, we formally show $|P_h^K| \leq \tilde{\mathcal{O}}(\varepsilon_S^{-z})$ (up to metric constants defined in Appendix A.2) by arguing that only near-optimal balls refine at fine scales, which we also use to achieve Eq. (6).

4.4 Key technical ideas

Several proof ingredients are new to the MDP setting and may be of independent interest.

Relaxation of exponential weights condition. To apply the varying action-set exponential weights lemma (Lemma 9, adapted from Podimata and Slivkins 2021), we must ensure $\eta_k |\hat{g}_k(B)| \leq 1$ on each ball, where $\hat{g}_k(B)$ is the estimated-Q minus dilated bonus (Line 5). The original lemma derives this from a condition connecting the exploration rate to the action-set size, which would imply the need for a layer-dependent exploration rate in our setting. We instead verify this condition directly using our estimator and the activation condition on the dilated bonus, allowing a global exploration rate $\gamma_k = \tilde{\mathcal{O}}(H\eta_k)$ that keeps the bonus sum at $\hat{V} = \tilde{\mathcal{O}}(H^{3/2} \sqrt{|P^K|K})$. See Remark 3 for details.

Connecting zoom-in rule and the zooming dimension. The zoom-in rule is triggered by confidence conditions – not (near-)optimality – so one must show that only near-optimal balls are refined below a radius threshold $\varepsilon_i = 1/\log K$. In particular, for balls with radius at

least $1/\log K$, the covering number of $\mathcal{S}_h \times \mathcal{A}$ is $\tilde{\mathcal{O}}(1)$, so all balls – regardless of optimality – contribute only a polylogarithmic additive term to $|P_h^K|$. For finer balls, the instantaneous confidence condition prevents any ball whose sampling probability is at the exploration floor from triggering zoom-in. Consequently, fine-scale balls can only arise as descendants of coarser near-optimal ancestors, whose near-optimality propagates to children via the $\mathcal{O}(LH)$ -Lipschitzness of Q^* . The full three-case argument is in Appendix A.5.

5 Conclusion

In this paper, we demonstrate that adaptive discretization and policy optimization can be made compatible in the continuous adversarial MDPs: a combination that has resisted prior efforts because adaptive zooming requires a time-varying action set, while dilated exploration bonuses are designed for stationary ones. By combining an instantaneous-confidence zooming rule to tame adversaries, dilated exploration bonuses to avoid occupancy measure mismatch, and a buffering mechanism to stabilize newly split balls, DiZOOM-PO achieves $\tilde{\mathcal{O}}(K^{(z+1)/(z+2)})$ regret where z is the adversarial zooming dimension. This improves over uniform discretization and matches the best adaptive rates for stochastic continuous MDPs, but instead in the strictly harder adversarial setting.

Several future research directions remain open. For example, extending our framework to richer function approximation classes, partially observable MDPs, or unknown metrics would further bridge the gap between the present theory and practical applications in reinforcement learning.

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A Appendix

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A.1 Related work

Learning in Adversarial MDPs. Online learning in adversarial MDPs was initiated by [Zimin and Neu \(2013\)](#), who cast the problem as online convex optimization over the polytope of valid occupancy measures. [Rosenberg and Mansour \(2019\)](#) extended this paradigm to unknown transitions via upper occupancy bounds, and [Jin et al. \(2020\)](#) further handled bandit feedback, establishing the $\tilde{\mathcal{O}}(\sqrt{K})$ regret now standard for the tabular occupancy-measure line. These methods require solving a convex program over a polytope whose size scales with the state-action cardinality each round, and their analyses rely on structural properties of this polytope that do not admit a direct continuous generalization. The *policy optimization* line ([Cai et al., 2020](#); [Shani et al., 2020](#); [Luo et al., 2021](#)) trades the global optimization for per-state local updates. [Shani et al. \(2020\)](#) established an initial $\tilde{\mathcal{O}}(K^{2/3})$ regret for tabular adversarial MDPs with bandit feedback, and [Luo et al. \(2021\)](#) subsequently attained $\tilde{\mathcal{O}}(\sqrt{K})$ by introducing dilated exploration bonuses that satisfy a modified Bellman equation, amplifying exploration at deeper layers to cancel the variance of adversarial importance-weighting estimators. Our work extends this technique from the tabular setting to continuous metric MDPs, which requires integrating it with a different discretization machinery. Yet another line utilizes the *Follow-the-Perturbed-Leader (FTPL)* framework, which is computationally easier but requires specialized analysis and thus is less

flexible comparatively (Neu et al., 2010, 2012, 2014; Wang and Dong, 2020; Dai et al., 2022; Chandrasekaran and Tewari, 2023).

Policy Optimization with Infinite States. Beyond the tabular case, a parallel line studies policy optimization under linear function approximation. Neu and Olkhovskaya (2021) gave the first sublinear-regret algorithm, achieving $\tilde{O}(K^{2/3})$ under an exploratory assumption. Luo et al. (2021) extended dilated bonuses to this setting, matching the rate without exploratory assumptions but requiring a simulator; without a simulator, their regret degrades to $\tilde{O}(K^{14/15})$. Subsequent works progressively improved the simulator-free rate: $\tilde{O}(K^{8/9})$ (Dai et al., 2023), $\tilde{O}(K^{6/7})$ (Sherman et al., 2023), $\tilde{O}(K^{4/5})$ under an exploratory assumption (Kong et al., 2024), and $\tilde{O}(K^{3/4})$ efficiently or $\tilde{O}(\sqrt{K})$ inefficiently (Liu et al., 2024). With full information, $\tilde{O}(\sqrt{K})$ is achievable efficiently (Sherman et al., 2024; Cassel and Rosenberg, 2024). The optimal efficient bandit-feedback algorithm remains open. All these works exploit the linear Q -function property that does not hold in our continuous metric setting; we instead rely on Lipschitz continuity, enabling adaptive discretization.

Continuous Metric MDPs. Continuous RL typically imposes a metric with Lipschitz-continuous optimal value functions (see, e.g., Kakade et al., 2003; Sinclair et al., 2019). Sinclair et al. (2019) gave an adaptive Q-learning algorithm for episodic metric MDPs whose regret scales with the worst-case covering dimension. Touati et al. (2020) and Cao and Krishnamurthy (2020) concurrently refined the analysis to scale with the zooming dimension of the instance, achieving the first instance-dependent rates for metric MDPs; the latter additionally introduced a buffering mechanism to stabilize updates on newly appearing partitions, which we adopt in our DiZOOM-PO. However, these existing methods estimate the stochastic and stationary rewards, and thus do not extend to adversarial losses. We are not aware of any prior regret guarantee for continuous adversarial MDPs beyond the naive uniform discretization (i.e., scaling with the covering rather than zooming dimension).

Adaptive Discretization and Adversarial Zooming. The “zooming” idea originated in Lipschitz bandits (Kleinberg, 2004; Bubeck et al., 2011; Kleinberg et al., 2019), where adaptive discretization captures the instance-dependent underlying geometry and achieves regret scaling with the zooming dimension (rather than the ambient covering dimension). Adaptive discretization in the adversarial setting was long open: standard discretization rule contains an average operation that an adversary can manipulate, causing the algorithm to over-refine uninformative regions. Podimata and Slivkins (2021) resolved this for adversarial Lipschitz bandits by introducing an *instantaneous* confidence rule alongside the standard one, with splitting triggered only when the current action’s probability is sufficiently large. We adopt their definition of the adversarial zooming dimension. We also define a similar instantaneous-confidence rule in multi-layer MDPs, thus allowing the discretization rule

to interleave with dilated bonuses and the buffering phase required for adversarial policy optimization.

A.2 Definitions of dimensions and constants

All definitions below are standard in the Lipschitz bandit/MDP literature (Kleinberg, 2004; Kleinberg et al., 2019; Podimata and Slivkins, 2021; Sinclair et al., 2019; Cao and Krishnamurthy, 2020).

Covering definitions. For a metric space $(\mathcal{Y}, D)$ and $r > 0$, an r -net of $\mathcal{Y}$ is a set $S \subset \mathcal{Y}$ such that the balls $\{B(y, r) : y \in S\}$ cover $\mathcal{Y}$. The covering number $N_r(\mathcal{Y})$ is the size of the smallest r -net of $\mathcal{Y}$. The doubling constant $C_{\text{dbl}} := \max_{y \in \mathcal{Y}, r > 0} N_{r/2}(B(y, r))$ is the largest number of radius- $r/2$ balls needed to cover any radius- r ball; for the d -dimensional unit cube under any ℓ_p metric, $C_{\text{dbl}} = 2^{\mathcal{O}(d)}$. For a given multiplier $\gamma > 0$, the γ -covering dimension of $\mathcal{S} \times \mathcal{A}$ is defined as

$$d := \inf\{d' \geq 0 : N_r(\mathcal{S} \times \mathcal{A}) \leq \gamma r^{-d'} \forall r > 0\}.$$

This depends only on the metric space and is therefore known to the algorithm.

Adversarial zooming dimension. Following Podimata and Slivkins (2021), the adversarial gap of state-action pair (x, a) at time t is, since the loss is being minimized,

$$\text{AdvGap}_t(x, a) := \frac{1}{t} \left[\sum_{\tau=1}^t Q_{\tau, h}^*(x, a) - \min_{(x', a')} \sum_{\tau=1}^t Q_{\tau, h}^*(x', a') \right],$$

i.e., the per-episode excess of (x, a) 's cumulative Q-value over that of the cumulative-best pair. We call (x, a) *inclusively ε -optimal* if $\text{AdvGap}_t(x, a) < \mathcal{O}(\varepsilon \log^{3/2} K)$ for some $t > \Omega(\varepsilon^{-2})$. Let $\mathcal{A}_\varepsilon$ denote the set of inclusively ε -optimal pairs. The adversarial γ -zooming dimension is

$$z := \inf\{z' \geq 0 : \mathcal{A}_\varepsilon \text{ can be covered with } \gamma \varepsilon^{-z'} \text{ sets of diameter } \leq \varepsilon, \forall \varepsilon > 0\}.$$

We always have $z \in [0, d]$, with $z = 0$ achievable on benign instances even when d is large.

Transition confidence radius. The radius $\text{rad}_h^k(B, B')$ used in Eq. (1) is the Bernstein-type confidence radius for the empirical mean transition estimate $\hat{P}_h^k(B' | B)$, namely

$$\text{rad}_h^k(B, B') := \sqrt{\frac{2\hat{P}_h^k(B' | B)(1 - \hat{P}_h^k(B' | B))\iota}{\max\{1, n_h^k(B)\}}} + \frac{2\iota}{3\max\{1, n_h^k(B)\}},$$

with $\iota := \log(|P^K|HK/\delta)$. A standard application of Lemma 5 with a union bound over all

h, B, B' guarantees that $\mathcal{P} \in \mathcal{P}_j$ for all j with probability at least $1 - \delta$.

Full form of Eq. (6) in Theorem 1.

$$\text{Reg}(K) = \tilde{\mathcal{O}} \left(L^{\frac{z}{z+2}} \cdot H^{\frac{3(z+1)}{z+2}} \cdot K^{\frac{z+1}{z+2}} \cdot (\gamma \cdot C_{\text{dbl}})^{\frac{1}{z+2}} \right). \quad (9)$$

A.3 Standard technical lemmas

Lemma 4 (Freedman's Inequality). *Let $(X_t)_{t=1}^T$ be a real-valued martingale difference sequence with respect to a filtration $(\mathcal{F}_t)_{t=0}^T$, where $(\mathcal{F}_k)_{k \geq 0}$ is the natural filtration generated by the interaction history up to the beginning of episode k and $\mathcal{F}_0$ is the trivial σ -algebra. Suppose there exists a constant $b > 0$ such that $|X_t| \leq b$ almost surely for all $t \in [T]$. Let $V_T = \sum_{t=1}^T \mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}]$ denote the sum of conditional variances. For any $\lambda \in (0, 1/b]$, with probability at least $1 - \delta$, we obtain*

$$\sum_{t=1}^T X_t \leq \lambda V_T + \frac{1}{\lambda} \ln \frac{1}{\delta}.$$

Lemma 5 (Bernstein's Inequality). *Let $X_1, \dots, X_n$ be independent random variables with $|X_i - \mathbb{E}[X_i]| \leq b$ almost surely. Let $S_n = \sum_{i=1}^n X_i$ and $\sigma^2 = \sum_{i=1}^n \text{Var}[X_i]$. Then for any $t > 0$,*

$$\mathbb{P}(|S_n - \mathbb{E}[S_n]| \geq t) \leq 2 \exp \left(-\frac{t^2}{2\sigma^2 + 2bt/3} \right).$$

Lemma 6 (Performance Difference Lemma (Kakade and Langford, 2002)). *For any two policies π, π' and any loss function ℓ ,*

$$V_1^\pi(x_1; \ell) - V_1^{\pi'}(x_1; \ell) = \sum_{h=1}^H \mathbb{E}_{x_h \sim d_h^\pi} \left[\mathbb{E}_{a \sim \pi(\cdot \mid x_h)} \left[Q_h^{\pi'}(x_h, a; \ell) \right] - V_h^{\pi'}(x_h; \ell) \right],$$

where d_h^π is the state distribution at layer h induced by π .

A.4 Proof of main guarantee Theorem 1

Proof of Theorem 1. By Lemma 3,

$$\text{Reg}(K) \leq \tilde{\mathcal{O}}(H/\eta_K) + 4 \sum_{k=1}^K \hat{V}_1^{\pi_k}(x_1; b_k + L \sum_{k,h,B} (q_{k,h}^{\star, \pi_k}(B) + q_{k,h}^{\star, \pi^*}(B)) r(B).$$

For a given radius threshold $\varepsilon_S > 0$, we decompose the discretization error by ball radius episode the threshold ε_S . Balls with $r(B) \leq \varepsilon_S$ contribute

$$L \sum_{k,h,B: r(B) \leq \varepsilon_S} (q_{k,h}^{\star, \pi_k}(B) + q_{k,h}^{\star, \pi^\star}(B)) r(B) \leq O(LH^3 K \varepsilon_S),$$

where the additional H^2 factor comes from the residual substitution in the dilated-bonus proof of Appendix A.7. Balls with $r(B) > \varepsilon_S$ keep at most their initial radius $r(B) = d_{\max}/H$ and contribute at most

$$\begin{aligned} L \sum_{k,h,B: r(B) > \varepsilon_S} (q_{k,h}^{\star, \pi_k}(B) + q_{k,h}^{\star, \pi^\star}(B)) \cdot \frac{d_{\max}}{H} \\ \leq L \cdot \frac{d_{\max}}{H} \cdot 2HK = 2LKd_{\max}, \end{aligned}$$

since the weights $q^{\star, \pi_k} + q^{\star, \pi^\star}$ sum to $2H$ across all balls per episode. It suffices to bound the bonus sum $\sum_k \widehat{V}_1^{\pi_k}(x_1; b_k)$, whose one-step bonus numerator is $3\gamma_k H + H(\bar{q}_k - \underline{q}_k) + (1 + 4 \log K)\beta_k$. The bonus numerator $3\gamma_k H + H(\bar{q}_k - \underline{q}_k) + \beta_k$ includes the $H(\bar{q} - \underline{q})$ term by design: indeed, the Bias-1 expected bias (Lemma 11) contributes $\sum q^{\star, \pi_k} \frac{H(\bar{q}_k - \underline{q}_k + \gamma_k)}{\bar{q}_k + \gamma_k}$ to the right-hand side, which $b_{k,h}$ incorporates, whence the residual bonus is

$$b_{k,h}(B) - \frac{H(\bar{q}_k - \underline{q}_k + \gamma_k)}{\bar{q}_k + \gamma_k} = \frac{2\gamma_k H + \beta_k}{\bar{q}_k + \gamma_k} \leq \frac{3\gamma_k H}{\bar{q}_k + \gamma_k},$$

which is nonnegative and depends only on γ_k , not on $\bar{q} - \underline{q}$. Hence the bonus-sum bound only needs to track the $\gamma_k H / (\bar{q}_k + \gamma_k)$ coefficient. The β_k contribution is $\tilde{\mathcal{O}}(1/\eta_K)$, also incorporated. For each layer h , the per-layer contribution satisfies

$$\sum_B q_{k,h}^{\pi_k}(B) \frac{\gamma_k H}{\bar{q}_k(B) + \gamma_k} \leq \gamma_k H \cdot |P_h^k|,$$

since each ball contributes at most γ_k to the sum $\sum_B q/(\bar{q} + \gamma_k)$ using $q/(\bar{q} + \gamma) \leq 1$. Summing over layers, the aggregate per-layer contribution is bounded by $\gamma_k H |P^k| = \tilde{\mathcal{O}}(\eta_k H^2 |P^k|)$. Summing over k with $\eta_k = (|P^k| HK)^{-1/2}$,

$$4H^2 \sum_{k=1}^K \eta_k |P^k| = 4H^2 \sum_{k=1}^K \frac{|P^k|}{\sqrt{|P^k| HK}} = 4H^{3/2} \frac{1}{\sqrt{K}} \sum_{k=1}^K \sqrt{|P^k|} \leq 4H^{3/2} \sqrt{|P^K| K}.$$

Adding the $\tilde{\mathcal{O}}(H/\eta_K) = \tilde{\mathcal{O}}(H^{3/2}\sqrt{|P^K|K})$ concentration remainder from Lemmas 11 and 12 of the same order, the total regret is

$$\text{Reg}(K) \leq \tilde{\mathcal{O}}(H^{3/2}\sqrt{|P^K|K} + LH^3K\varepsilon_S). \quad (10)$$

By Lemma 7 below, $|P_h^K| \leq \tilde{\mathcal{O}}(C_{\text{dbl}}\gamma\varepsilon_S^{-z})$ for every $h \in [H]$, hence $|P^K| \leq H \cdot \tilde{\mathcal{O}}(C_{\text{dbl}}\gamma\varepsilon_S^{-z})$. Substituting into Eq. (10),

$$\text{Reg}(K) \leq \tilde{\mathcal{O}}(H^{3/2}\sqrt{HK \cdot C_{\text{dbl}}\gamma\varepsilon_S^{-z}} + LH^3K\varepsilon_S).$$

Selecting the threshold ε_S as

$$\varepsilon_S := (\sqrt{C_{\text{dbl}}\gamma}/(LH^{3/2}\sqrt{K}))^{2/(z+2)}$$

produces Eq. (9), as desired. $\square$

A.5 Bound on the total number of active nalls

We extract the bound $|P_h^K| \leq \tilde{\mathcal{O}}(C_{\text{dbl}}\gamma\varepsilon_S^{-z})$ used in Appendix A.4 into a separate lemma.

Lemma 7 (Active-ball count). *For every $h \in [H]$ and every $\varepsilon_S \in (0, 1]$,*

$$|P_h^K| \leq \tilde{\mathcal{O}}(C_{\text{dbl}}\gamma\varepsilon_S^{-z}).$$

Proof. The zoom-in rule activates only on balls already in the active set, and the buffering condition requires $N_{\min}(B) = \lceil r(B)^{-2}/4 \rceil$ visits before promotion. Since the parent of B remains active during the buffering phase, every play of a parent at layer h increments the visit counter for itself together with its at most C_{dbl} buffering children. Each of the K episodes contributes one play per layer, whence the total number of visit-counter increments across all episodes, layers, and balls is at most $(1 + C_{\text{dbl}})KH$. Since each ball B that has been zoomed must have accumulated at least $r(B)^{-2}/4$ of these increments,

$$\sum_{\text{zoomed } B} r(B)^{-2} \leq 4(1 + C_{\text{dbl}})KH, \quad (11)$$

where the sum is over the set of all zoomed balls.

We partition the set of all zoomed balls into geometric scales corresponding to radii $2^{-i}d_{\max}$ for $i \in \{0, 1, \dots, \mathcal{O}(\log K)\}$, and group balls into sets where the radius is in $(2^{-(i+1)}d_{\max}, 2^{-i}d_{\max}]$. Recall from the algorithm that whenever a ball B triggers zoom-

in at episode k , the instantaneous condition $\text{conf}_k^{\text{inst}}(B) \leq e^{r(B)} - 1$ is satisfied. Substituting $\text{conf}_k^{\text{inst}}(B) = \eta_k + \eta_k/\pi_k(B \mid x_B)$ and rearranging gives

$$\pi_k(B \mid x_B) \geq \frac{\eta_k}{e^{r(B)} - 1 - \eta_k}. \quad (12)$$

Because every ball $B \in \text{rel}_h^k(x)$ receives at least the exploration floor $\pi_k(B \mid x) \geq \gamma_k/|\text{rel}_h^k(x)| \geq \gamma_k/|P_h^k| = (2 + 4 \log K) \cdot 2H\eta_k/|P_h^k|$, the condition (12) can be satisfied at the floor only when $r(B) \geq \ln(1 + |P_h^k|/((2 + 4 \log K) \cdot 2H)) \geq 1/(4 \log K)$ (using $|P_h^k| \geq 1$). Hence balls at scales $r(B) < 1/(4 \log K)$ can only zoom in when the exponential weights component $\tilde{\pi}_k$ assigns B nontrivial weight beyond the exploration floor.

We now partition the scales into three ranges and bound the zoomed balls at each.

Case 1: $2^{-i}d_{\max} > \log K$. At these scales, the covering number $N_{2^{-i}d_{\max}}(\mathcal{S} \times \mathcal{A})$ is $\tilde{\mathcal{O}}(1)$ (since the diameter $d_{\max}$ is bounded and $2^{-i}d_{\max}$ itself is bounded above by $d_{\max}$, and r is bounded below by $\log K$ on this range). Hence the total number of balls at coarse scales is $\tilde{\mathcal{O}}(1)$, regardless of optimality.

Case 2: $1/\log K \leq 2^{-i}d_{\max} \leq \log K$. At any fixed scale in this range, the number of nonoverlapping balls that can co-exist in $\mathcal{S}_h \times \mathcal{A}$ is $\tilde{\mathcal{O}}(1)$, since the covering dimension d contributes only a $\text{polylog}(K)$ factor. Hence at most $\tilde{\mathcal{O}}(1)$ balls per layer can exist at these scales, regardless of optimality.

Case 3: $2^{-i}d_{\max} < 1/\log K$. At these scales, zoom-in cannot occur for any ball at the exploration floor (by Eq. (12)). Hence fine-scale balls come only from two sources: (a) descendants of coarser-scale near-optimal parents, whose near-optimality propagates via Lipschitzness (since $\text{AdvGap}(\text{child}) \leq \text{AdvGap}(\text{parent}) + \mathcal{O}(LHr(\text{parent}))$) by the $\mathcal{O}(LH)$ -Lipschitzness of Q^* ; and (b) descendants of the at most $\tilde{\mathcal{O}}(1)$ intermediate scale parents, which may be suboptimal. Balls in case (a) satisfy the condition that $2^{-i}d_{\max}$ is inclusively-optimal, hence the number of such balls is at most $\gamma(2^{-i}d_{\max})^{-z}$ by definition of the zooming dimension. Balls in case (b) are independently capped by the visit-count (11): each ball at scale $2^{-i}d_{\max}$ requires $(2^{-i}d_{\max})^{-2}$ visits, and the total limits the count at each scale to $\mathcal{O}(KH(2^{-i}d_{\max})^2)$, which is roughly $\gamma(2^{-i}d_{\max})^{-z}$ at fine scales.

Each split operation generates at most C_{dbl} children. Combining the three cases,

$$|P_h^K| \leq C_{\text{dbl}} \sum_{i=0}^{\mathcal{O}(\log K)} \gamma(2^{-i}d_{\max})^{-z} + \tilde{\mathcal{O}}(1). \quad (13)$$

The geometric sum evaluates as

$$\sum_{i=0}^{\mathcal{O}(\log K)} (2^{-i} d_{\max})^{-z} = d_{\max}^{-z} \sum_{i=0}^{\mathcal{O}(\log K)} 2^{iz} < \frac{2^z}{2^z - 1} \varepsilon_S^{-z},$$

hence $|P_h^K| \leq \frac{2^z C_{\text{dbl}} \gamma}{2^z - 1} \varepsilon_S^{-z} = \tilde{\mathcal{O}}(C_{\text{dbl}} \gamma \varepsilon_S^{-z})$, as desired. $\square$

A.6 Proof of continuous-to-discrete reduction Lemma 3

Proof of Lemma 3. By the performance difference Lemma 6, the total regret satisfies $\text{Reg}(K) = \sum_{k=1}^K \sum_{h=1}^H \mathbb{E}_{x_h \sim q_{k,h}^*} [\dots]$ such that the continuous state performance difference contribution for each layer and episode is precisely

$$\mathbb{E}_{x_h \sim q_{k,h}^*} \left[\sum_{B \in \text{rel}_h^k(x_h)} \pi_k(B \mid x_h) Q_{k,h}^{\pi_k}(x_h, a_{\pi_k}(x_h, B)) - \mathbb{E}_{a \sim \pi^*(\cdot \mid x_h)} [Q_{k,h}^{\pi_k}(x_h, a)] \right].$$

The corresponding ball-level analogue is

$$\sum_{B \in P_h^k} (q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B)) Q_{k,h}^{\pi_k}(B),$$

where recall $q_{k,h}^{\star, \pi^*}(B) = \mathbb{E}_{x_h \sim q_{k,h}^*} [\Pr_{a \sim \pi^*(\cdot \mid x_h)} ((x_h, a) \in B)]$. We aim to establish that the continuous state performance difference term is bounded by the ball-level analogue plus the discretization errors:

$$\begin{aligned} & \mathbb{E}_{x_h \sim q_{k,h}^*} \left[\sum_{B \in \text{rel}_h^k(x_h)} \pi_k(B \mid x_h) Q_{k,h}^{\pi_k}(x_h, a_{\pi_k}(x_h, B)) - \mathbb{E}_{a \sim \pi^*(\cdot \mid x_h)} [Q_{k,h}^{\pi_k}(x_h, a)] \right] \\ & \leq \sum_{B \in P_h^k} (q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B)) Q_{k,h}^{\pi_k}(B) \\ & \quad + L \sum_B q_{k,h}^{\star, \pi_k}(B) r(B) + L \sum_B q_{k,h}^{\star, \pi^*}(B) r(B). \end{aligned} \tag{14}$$

We bound each contribution to the difference between the continuous and ball-level performance differences by virtue of the Lipschitz Assumption 1. For the π_k -weighted contribution, since $(x_h, a_{\pi_k}(x_h, B)) \in B$ in tandem with $c(B)$ being the center of B at distance at most $r(B)$, Assumption 1 delivers the upper bound $Q_{k,h}^{\pi_k}(x_h, a_{\pi_k}(x_h, B)) \leq Q_{k,h}^{\pi_k}(B) + Lr(B)$. Multiplying

by $\pi_k(B \mid x_h)$, summing over $B \in \text{rel}_h^k(x_h)$, and taking expectation under $x_h \sim q_{k,h}^*$,

$$\mathbb{E}_{x_h} \left[\sum_B \pi_k(B \mid x_h) Q_{k,h}^{\pi_k}(x_h, a_{\pi_k}(x_h, B)) \right] \leq \sum_B q_{k,h}^{\star, \pi_k}(B) (Q_{k,h}^{\pi_k}(B) + Lr(B)).$$

For the π^* -weighted contribution, the partition P_h^k covers $\mathcal{S}_h \times \mathcal{A}$, whence every action $a \sim \pi^*(\cdot \mid x_h)$ lies in some ball $B \in P_h^k$. Invoking the Lipschitz lower bound $Q_{k,h}^{\pi_k}(x_h, a) \geq Q_{k,h}^{\pi_k}(B) - Lr(B)$ for the ball B containing (x_h, a) ,

$$\mathbb{E}_{a \sim \pi^*(\cdot \mid x_h)} [Q_{k,h}^{\pi_k}(x_h, a)] \geq \sum_B \Pr_{a \sim \pi^*(\cdot \mid x_h)} ((x_h, a) \in B) (Q_{k,h}^{\pi_k}(B) - Lr(B)).$$

Taking expectation over $x_h \sim q_{k,h}^*$ in tandem with $\mathbb{E}_{x_h} [\Pr_{a \sim \pi^*(\cdot \mid x_h)} ((x_h, a) \in B)] = q_{k,h}^{\star, \pi^*}(B)$ establishes Eq. (14).

Summing Eq. (14) over k and h and invoking Lemma 2 to bound the ball-level analogue, we find that

$$\text{Reg}(K) \leq \tilde{\mathcal{O}}(H/\eta_K) + 4 \sum_{k=1}^K \hat{V}_1^{\pi_k}(x_1; b_k) + L \sum_{k,h,B} (q_{k,h}^{\star, \pi_k}(B) + q_{k,h}^{\star, \pi^*}(B)) r(B),$$

which is the claim once the leading bonus term is identified with the ball-level regret in the statement of Lemma 3, as desired. $\square$

Remark 2. *The proof of Lemma 3 uses Lipschitzness of Q^{π_k} , not only Q^* , which is why Assumption 1 is stated for all policies. This is a mild condition: Q^π inherits Lipschitzness whenever the transition kernel and loss functions are Lipschitz, as these properties propagate through the Bellman recursion. Attempting to bypass per-policy Lipschitzness by decomposing $Q^{\pi_k} - Q^*$ and bounding the residual via the suboptimality gap creates a recursive coupling between the discretization error and the regret itself.*

A.7 Proof of ball-level dilated-bonus cancellation Lemma 2

We now prove a generalized version of Lemma 2.

Lemma 8 (Ball-level dilated-bonus lemma; analogue of Lemma B.1 in Luo et al. 2021). *Suppose the dilated bonus $\mathcal{B}_{k,h}(B)$ satisfies, for all k, h, B ,*

$$\mathcal{B}_{k,h}(B) = b_{k,h}(B) + \left(1 + \frac{1}{H}\right) \max_{\hat{P} \in \mathcal{P}_j} \sum_{B'} \hat{P}(B' \mid B) \sum_{C \in \text{rel}_{h+1}^k(x_{B'})} \pi_k(C \mid x_{B'}) \mathcal{B}_{k,h+1}(C) \quad (15)$$

$$\geq b_{k,h}(B) + \left(1 + \frac{1}{H}\right) \sum_{B'} P_h(B' | B) \sum_C \pi_k(C | x_{B'}) \mathcal{B}_{k,h+1}(C),$$

with $\mathcal{B}_{k,H+1} \equiv 0$, and suppose that Eq. (28) holds. We obtain

$$\sum_{k=1}^K \sum_{h=1}^H \sum_B \left(q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B) \right) Q_{k,h}^{\pi_k}(B) \leq \tilde{\mathcal{O}}(H/\eta_K) + 4 \sum_{k=1}^K \hat{V}_1^{\pi_k}(x_1; b_k), \quad (16)$$

where $\hat{V}^\pi(\cdot; b_k)$ is the state value function under the optimistic transition $\hat{P}^k$ and policy π .

Proof. By Eq. (28), the ball-level regret is bounded by $\tilde{\mathcal{O}}(H/\eta_K)$ plus the sums:

$$\begin{aligned} & \sum_{k,h,B} q_{k,h}^{\star, \pi_k}(B) b_{k,h}(B) + \sum_{k,h,B} q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B) \\ & + \frac{1}{H} \sum_{k,h,B} q_{k,h}^{\star, \pi_k}(B) \mathcal{B}_{k,h}(B) + \sum_{k,h,B} (q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B)) \mathcal{B}_{k,h}(B). \end{aligned}$$

Our strategy is to telescope the latter two sums across layers via the lower-bound recursion (15), after which the $q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B)$ sum inherited from BIAS-2 cancels with a matching negative term produced by the telescoping.

We first establish a uniform bound on the dilated bonus. Since

$$b_{k,h}(B) \leq 3H + H(\bar{q}_k - \underline{q}_k)/(\bar{q}_k + \gamma_k) + (1 + 4 \log K) \beta_k/(\bar{q}_k + \gamma_k) \leq 5H$$

by virtue of $(\bar{q} - \underline{q})/(\bar{q} + \gamma) \leq 1$ and $\beta_k/\gamma_k \leq 1/(2H)$, in tandem with

$$\sum_{B'} \hat{P}_h^k(B' | B) \sum_C \pi_k(C | x_{B'}) = 1,$$

backward induction on the recursion (15) delivers $\mathcal{B}_{k,h}(B) \leq 5H \sum_{j=0}^{H-h} (1 + 1/H)^j \leq 5eH^2$ for all k, h, B .

We now telescope: fix an episode k and a layer h . Expanding $\mathcal{B}_{k,h}(B)$ via the lower bound in Eq. (15) with the true transition P_h , and using the Lipschitz Assumption 1 total-variation bound $\|P_h(\cdot | x, a) - P_h(\cdot | c(B))\|_{\text{TV}} \leq Lr(B)$ for $(x, a) \in B$, we bound the difference incurred by swapping the ball center $c(B)$ with the true state-action pair directly:

$$\begin{aligned} & \sum_B q_{k,h}^{\star, \pi^*}(B) \mathcal{B}_{k,h}(B) \\ & \geq \sum_B q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B) + \left(1 + \frac{1}{H}\right) \sum_C q_{k,h+1}^{\star, \pi_k}(C) \mathcal{B}_{k,h+1}(C) \end{aligned}$$

$$-10eLH^2 \sum_B q_{k,h}^{\star, \pi^*}(B) r(B).$$

Rearranging, we have

$$\begin{aligned} & \sum_B (q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B)) \mathcal{B}_{k,h}(B) \\ & \leq \sum_B q_{k,h}^{\star, \pi_k}(B) \mathcal{B}_{k,h}(B) - \sum_B q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B) \\ & \quad - \left(1 + \frac{1}{H}\right) \sum_C q_{k,h+1}^{\star, \pi_k}(C) \mathcal{B}_{k,h+1}(C) + 10eLH^2 \sum_B q_{k,h}^{\star, \pi^*}(B) r(B). \end{aligned} \quad (17)$$

Summing Eq. (17) over $h \in [H]$ and adding the sum scaled by $1/H$, the $(1+1/H) \sum q_{k,h}^{\star, \pi_k}(B) \mathcal{B}_{k,h}(B)$ contribution at layer h cancels against the $-(1+1/H) \sum q_{k,h+1}^{\star, \pi_k}(C) \mathcal{B}_{k,h+1}(C)$ contribution at layer $h+1$. By virtue of $\mathcal{B}_{k,H+1} \equiv 0$ terminating the telescoping at the boundary,

$$\begin{aligned} & \sum_{h=1}^H \sum_B \left[\frac{1}{H} q_{k,h}^{\star, \pi_k}(B) \mathcal{B}_{k,h}(B) + (q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B)) \mathcal{B}_{k,h}(B) \right] \\ & \leq \left(1 + \frac{1}{H}\right) \sum_B q_{k,1}^{\star, \pi_k}(B) \mathcal{B}_{k,1}(B) - \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B) \\ & \quad + 10eLH^2 \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi^*}(B) r(B). \end{aligned}$$

The residual sum satisfies $10eLH^2 \sum_{h,B} q_{k,h}^{\star, \pi^*}(B) r(B) \leq 10eLH^2 \cdot H\varepsilon_S = 10eLH^3\varepsilon_S$ per episode, which is incorporated into the discretization error of Lemma 3, strengthening $2LHK\varepsilon_S$ to $O(LH^3K\varepsilon_S)$. The $q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B)$ sum cancels exactly when adding it to both sides, whence adding the $q_{k,h}^{\star, \pi_k}(B) b_{k,h}(B)$ sum yields

$$\begin{aligned} & \sum_{h=1}^H \sum_B \left[q_{k,h}^{\star, \pi_k}(B) b_{k,h}(B) + q_{k,h}^{\star, \pi^*}(B) b_{k,h}(B) \right. \\ & \quad \left. + \frac{1}{H} q_{k,h}^{\star, \pi_k}(B) \mathcal{B}_{k,h}(B) + (q_{k,h}^{\star, \pi_k}(B) - q_{k,h}^{\star, \pi^*}(B)) \mathcal{B}_{k,h}(B) \right] \\ & \leq \left(1 + \frac{1}{H}\right) \sum_B q_{k,1}^{\star, \pi_k}(B) \mathcal{B}_{k,1}(B) + \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) b_{k,h}(B). \end{aligned} \quad (18)$$

We now bound the layer-1 residual on the right-hand side of Eq. (18) by a backward induction on the layer index h . The inductive claim is that, for every $h \in [H]$ and every state

$x \in \mathcal{S}_h$,

$$\sum_{B \in \text{rel}_h^k(x)} \pi_k(B | x) \mathcal{B}_{k,h}(B) \leq \left(1 + \frac{1}{H}\right)^{H-h} \widehat{V}_h^{\pi_k}(x; b_k). \quad (19)$$

The base case $h = H$ is immediate: by virtue of the recursion (15) in tandem with $\mathcal{B}_{k,H+1} \equiv 0$, we have $\mathcal{B}_{k,H}(B) = b_{k,H}(B)$, whence $\sum_B \pi_k(B | x) \mathcal{B}_{k,H}(B) = \widehat{V}_H^{\pi_k}(x; b_k)$. For the inductive step, suppose Eq. (19) holds at layer $h + 1$. Expanding via Eq. (15) with the optimistic transition $\widehat{P}_h^k$ and invoking the inductive hypothesis on the inner sum,

$$\begin{aligned} & \sum_B \pi_k(B | x) \mathcal{B}_{k,h}(B) \\ &= \sum_B \pi_k(B | x) \left[b_{k,h}(B) + \left(1 + \frac{1}{H}\right) \sum_{B'} \widehat{P}_h^k(B' | B) \sum_C \pi_k(C | x_{B'}) \mathcal{B}_{k,h+1}(C) \right] \\ &\leq \sum_B \pi_k(B | x) \left[b_{k,h}(B) + \left(1 + \frac{1}{H}\right)^{H-h} \sum_{B'} \widehat{P}_h^k(B' | B) \widehat{V}_{h+1}^{\pi_k}(x_{B'}; b_k) \right] \\ &\leq \left(1 + \frac{1}{H}\right)^{H-h} \widehat{V}_h^{\pi_k}(x; b_k), \end{aligned}$$

where the second inequality relies on the inductive hypothesis and the third on $b_{k,h} \geq 0$ in tandem with the recursive Bellman identity for $\widehat{V}$. The induction is complete, establishing Eq. (19).

The deterministic initial state x_1 implies that

$$\sum_B q_{k,1}^{\star, \pi_k}(B) \mathcal{B}_{k,1}(B) = \sum_{B \in \text{rel}_1^k(x_1)} \pi_k(B | x_1) \mathcal{B}_{k,1}(B),$$

whence Eq. (19) at $h = 1$ delivers

$$\begin{aligned} \left(1 + \frac{1}{H}\right) \sum_B \pi_k(B | x_1) \mathcal{B}_{k,1}(B) &\leq \left(1 + \frac{1}{H}\right)^H \widehat{V}_1^{\pi_k}(x_1; b_k) \\ &\leq 3 \widehat{V}_1^{\pi_k}(x_1; b_k), \end{aligned}$$

by virtue of $(1 + 1/H)^H \leq e < 3$ for $H \geq 1$. The residual $\sum_h \sum_B q_{k,h}^{\star, \pi_k}(B) b_{k,h}(B)$ in Eq. (18) is bounded by $\widehat{V}_1^{\pi_k}(x_1; b_k)$ via the same backward induction applied to the bonus value function. Summing Eq. (18) over $k \in [K]$,

$$\text{Reg}(K) \leq (3 + 1) \sum_{k=1}^K \widehat{V}_1^{\pi_k}(x_1; b_k) = 4 \sum_{k=1}^K \widehat{V}_1^{\pi_k}(x_1; b_k),$$

as desired. $\square$

A.8 Three-term decomposition: bounding Bias-1, Bias-2, and Reg-Term

We now prove the bounds on BIAS-1, BIAS-2, and REG-TERM that together yield the ball-level analogue of Eq. (5) in Luo et al. (2021), which feeds into the dilated-bonus cancellation in Appendix A.7.

A.8.1 Varying Action-Set Multiplicative Weights and Reg-Term

The dynamic variation of the active set of balls over time guarantees that standard multiplicative weights results do not apply. We proceed to employ a varying action-set multiplicative-weights lemma. The zooming of a ball guarantees that its weight is split equally among its children, an effect tracked by the term $W(B_K^*)$ in the lemma below.

Lemma 9 (Varying action-set exponential weights; adapted from Podimata and Slivkins (2021, Lemma C.7)). *Fix a layer $h \in [H]$ and a state $x \in \mathcal{S}_h$. Assume the sequences $\{\eta_k\}_{k=1}^K$ and $\{\beta_k\}_{k=1}^K$ are nonincreasing with $\eta_k \leq \beta_k$, and that for every $k \in [K]$ and every active ball $B \in \text{rel}_h^k(x)$,*

$$\eta_k |\hat{g}_k(B)| \leq 1. \quad (20)$$

Let $\hat{g}_k: \text{rel}_h^k(x) \rightarrow \mathbb{R}$ be estimated losses. The execution of exponential weights over $\text{rel}_h^k(x)$ with mixing probability γ_k guarantees that, for any comparator $B_K^ \in P_h^K$,*

$$\begin{aligned} & \sum_{k=1}^K \hat{g}_k(\text{anc}_k(B_K^*)) - \sum_{k=1}^K \sum_{B \in \text{rel}_h^k(x)} \pi_k(B | x) \hat{g}_k(B) \\ & \leq \frac{\ln(|\text{rel}_h^K(x)| W(B_K^*))}{\eta_K} + 4(1 + \log K) \sum_{k=1}^K \beta_k |\text{rel}_h^k(x)| \\ & \quad + 2 \sum_{k=1}^K \eta_k \left(1 + (1 + 4 \log K) \beta_k\right) \sum_{B \in \text{rel}_h^k(x)} \pi_k(B | x) \hat{g}_k(B)^2, \end{aligned} \quad (21)$$

where $\text{anc}_k(B_K^*)$ denotes the ancestor of B_K^* active at episode k and $W(B_K^*) = \prod_v |C(v)|$ over all ancestors v of B_K^* in the zooming tree.

Remark 3. The original Podimata and Slivkins (2021, Lemma C.7) assumes $\eta_t \leq \beta_t \leq \gamma_t/|A_t|$ and $\eta_t(1 + \beta_t(1 + 4 \log T)) \leq \gamma_t/|A_t|$. These conditions serve two purposes in their proof:

(i) deriving the precondition $\eta_t |\widehat{g}_t(u)| \leq 1$ (their Eq. (C.28)), and (ii) converting the per-episode exploration penalty $\beta_t |A_t|$ into γ_t (their Eq. (C.30)). In our setting, purpose (i) is fulfilled directly by Eq. (20), which we verify in the proof of Lemma 10. For purpose (ii), we keep $\beta_k |\text{rel}_h^k(x)|$ instead of γ_k in the exploration-cost term; the proof is otherwise line-for-line identical to that of Podimata and Slivkins (2021). Under our parameter choices $\beta_k = \eta_k$ and $\gamma_k = (2+4 \log K) \cdot 2H\eta_k$,

$$4(1+\log K) \sum_{k=1}^K \beta_k |\text{rel}_h^k(x)| \leq 4(1+\log K) \sum_{k=1}^K \eta_k |P_h^k| \leq \widetilde{\mathcal{O}} \left(\sqrt{|P^K|K/H} \right),$$

which is $O(1/H^2)$ times the leading $H/\eta_K = \widetilde{\mathcal{O}}(H^{3/2} \sqrt{|P^K|K})$ term and is therefore incorporated.

The buffering mechanism guarantees that balls are only included in the active set P_h^k once sufficiently explored. We require for all k, h and all $B \in P_h^k$ that

$$\eta_k \mathcal{B}_{k,h}(B) \leq \frac{1}{2H}. \quad (22)$$

Lemma 10. *Under the conditions above, the optimization regret REG-TERM satisfies*

$$\text{REG-TERM} \leq \widetilde{\mathcal{O}} \left(\frac{H}{\eta_K} \right) + \sum_{k=1}^K \sum_{h=1}^H \sum_B q_{k,h}^{*,\pi_k}(B) \left(\frac{\gamma_k H}{\bar{q}_k(B) + \gamma_k} + \frac{\mathcal{B}_{k,h}(B)}{H} \right). \quad (23)$$

Proof. Fix layer $h \in [H]$ and state $x \in \mathcal{S}_h$. The algorithm runs exponential weights over $\text{rel}_h^k(x) \subseteq P_h^k$ with loss $\widehat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B)$ for $B \in \text{rel}_h^k(x) \subseteq P_h^k$ using the exponential weights distribution $\pi_k(\cdot | x)$. We verify the condition of Lemma 9 and bound each term in the resulting regret estimate.

Concerning the condition $\eta_k |\widehat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B)| \leq 1$, the triangle inequality decomposes it into two pieces:

$$\eta_k |\widehat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B)| \leq \eta_k \widehat{Q}_{k,h}(B) + \eta_k \mathcal{B}_{k,h}(B) \leq \frac{1}{2} + \frac{1}{2H} \leq 1,$$

where the first piece relies on $\widehat{Q}_{k,h}(B) \leq H/\gamma_k$ in tandem with $\eta_k \leq \gamma_k/(2H)$ (since $\gamma_k \geq 2H\eta_k$), and the second piece is precisely the activation condition $\eta_k \mathcal{B}_{k,h}(B) \leq 1/(2H)$ in Eq. (22). Invoking Lemma 9 on the loss sequence then delivers, for any comparator $B^* \in \text{rel}_h^K(x)$,

$$\sum_{k=1}^K \sum_{B \in \text{rel}_h^k(x)} (\pi_k(B | x) - \pi^*(B | x)) (\widehat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B))$$

$$\begin{aligned}
&\leq \frac{\ln(|P_h^K|W(B^*))}{\eta_K} \\
&\quad + \eta_K \sum_{k=1}^K \sum_{B \in \text{rel}_h^k(x)} \pi_k(B | x) (\widehat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B))^2.
\end{aligned} \tag{24}$$

It remains to control the squared-loss term on the right-hand side. The elementary inequality $(a - b)^2 \leq 2a^2 + 2b^2$ delivers

$$\sum_B \pi_k(B | x) (\widehat{Q}_{k,h}(B) - \mathcal{B}_{k,h}(B))^2 \leq 2 \sum_B \pi_k(B | x) \widehat{Q}_{k,h}(B)^2 + 2 \sum_B \pi_k(B | x) \mathcal{B}_{k,h}(B)^2,$$

and we treat the two contributions in turn. For the bonus contribution, the activation condition Eq. (22) implies that $\mathcal{B}_{k,h}(B) \leq 1/(2\eta_k H)$, whence $\eta_k \mathcal{B}_{k,h}(B)^2 \leq \mathcal{B}_{k,h}(B)/(2H)$, so that

$$2\eta_k \sum_B \pi_k(B | x) \mathcal{B}_{k,h}(B)^2 \leq \frac{1}{H} \sum_B \pi_k(B | x) \mathcal{B}_{k,h}(B).$$

For the estimator contribution, the indicator $\mathbf{1}\{B = B_h^k\}$ inside $\widehat{Q}_{k,h}(B)$ admits at most one nonzero summand, and combined with $L_{k,h} \leq H$ we obtain $\widehat{Q}_{k,h}(B)^2 \leq H^2/(\bar{q}_k(B) + \gamma_k)^2 \cdot \mathbf{1}\{B = B_h^k\}$. Taking conditional expectation given the history prior to episode k (denoted by $\mathbb{E}_{k-1}[\cdot]$), since $\mathbb{E}_{k-1}[\mathbf{1}\{B = B_h^k\}] = q_k(B)$ and $q_k(B) \leq \bar{q}_k(B) + \gamma_k$,

$$\begin{aligned}
\mathbb{E}_{k-1} \left[\sum_B \pi_k(B | x) \widehat{Q}_{k,h}(B)^2 \right] &= \sum_B \pi_k(B | x) \frac{H^2 q_k(B)}{(\bar{q}_k(B) + \gamma_k)^2} \\
&\leq \sum_B \pi_k(B | x) \frac{H^2}{\bar{q}_k(B) + \gamma_k}.
\end{aligned}$$

Multiplying Eq. (24) by $q_{k,h}^*(x)$, summing over states $x \in \mathcal{S}_h$ and layers $h \in [H]$, and substituting the bounds above, by virtue of $\gamma_k \geq 2H\eta_k$ the inequality $2\eta_k H^2 \leq \gamma_k H$ ensures the estimator second moment is incorporated into the bonus-shaped term $\gamma_k H/(\bar{q}_k(B) + \gamma_k)$. The residual $\widetilde{\mathcal{O}}(H/\eta_K)$ subsumes the $\ln(|P_h^K|W(B^*))/\eta_K$ leading term in tandem with the polylogarithmic factors arising from the count of episodes and states, whence

$$\text{REG-TERM} \leq \widetilde{\mathcal{O}} \left(\frac{H}{\eta_K} \right) + \sum_{k=1}^K \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \left(\frac{\gamma_k H}{\bar{q}_k(B) + \gamma_k} + \frac{\mathcal{B}_{k,h}(B)}{H} \right),$$

which is Eq. (23). □

A.8.2 Bounding the Underestimation and Overestimation Biases (Bias-1 and Bias-2)

Lemma 11. *With probability at least $1 - 5\delta$, we obtain*

$$\text{BIAS-1} \leq \tilde{\mathcal{O}}\left(\frac{1}{\eta_K}\right) + \sum_{k=1}^K \sum_{h=1}^H \sum_{B \in P_h^k} q_{k,h}^{\star, \pi_k}(B) \frac{2\gamma_k H + H(\bar{q}_k(B) - \underline{q}_k(B))}{\bar{q}_k(B) + \gamma_k}.$$

Proof. Adding and subtracting the per-episode conditional expectation given the history prior to episode k (denoted by $\mathbb{E}_{k-1}[\cdot]$) across all layers simultaneously,

$$\begin{aligned} \text{BIAS-1} = & \underbrace{\sum_{k=1}^K \left(\mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \hat{Q}_{k,h}(B) \right] - \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \hat{Q}_{k,h}(B) \right)}_{\text{martingale}} \\ & + \underbrace{\sum_{k=1}^K \left(\sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) Q_{k,h}^{\pi_k}(B) - \mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \hat{Q}_{k,h}(B) \right] \right)}_{\text{expected bias}}. \end{aligned} \quad (25)$$

The point of summing over h before taking conditional expectations is that, although the per-layer differences do not form a martingale across the (k, h) index (the indicator $\mathbf{1}\{B = B_h^k\}$ at layer h is not measurable with respect to a per-step filtration that has already revealed layer $h + 1$), the per-episode aggregates form a martingale difference sequence with respect to the natural filtration.

For the martingale piece, an application of Lemma 4 requires a magnitude bound on the per-episode aggregate in tandem with a conditional second moment. The indicator $\mathbf{1}\{B = B_h^k\}$ inside $\hat{Q}_{k,h}(B)$ leaves at most one summand nonzero per layer, and combined with $q_{k,h}^{\star, \pi_k}(B) \leq 1$ and $L_{k,h} \leq H$, the per-layer sum is bounded by H/γ_k , whence the per-episode aggregate is bounded by $H^2/\gamma_{\min}$. Since $\sum_B q_{k,h}^{\star, \pi_k}(B) = 1$ at each layer, in tandem with the elementary variance bound,

$$\begin{aligned} & \mathbb{E}_{k-1} \left[\left(\sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \hat{Q}_{k,h}(B) - \mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \hat{Q}_{k,h}(B) \right] \right)^2 \right] \\ & \leq H \sum_{h=1}^H \mathbb{E}_{k-1} \left[\left(\sum_B q_{k,h}^{\star, \pi_k}(B) \hat{Q}_{k,h}(B) \right)^2 \right] \\ & \leq H \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \frac{H^2}{\bar{q}_k(B) + \gamma_k}, \end{aligned}$$

where the final step relies on $q_k(B) \leq \bar{q}_k(B) + \gamma_k$ together with $L_{k,h} \leq H$. Invoking Lemma 4 with $b = H^2/\gamma_{\min}$ and $\lambda = \gamma_{\min}/H^2$ (so that $\lambda b = 1$, validating the Freedman precondition $\lambda \in (0, 1/b]$) produces, with probability at least $1 - \delta$,

$$\begin{aligned} & \sum_{k=1}^K \left(\mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \widehat{Q}_{k,h}(B) \right] - \sum_{h=1}^H \sum_B q_{k,h}^{\star, \pi_k}(B) \widehat{Q}_{k,h}(B) \right) \\ & \leq \gamma_{\min} \sum_{k,h,B} q_{k,h}^{\star, \pi_k}(B) \frac{H}{\bar{q}_k(B) + \gamma_k} + \frac{H^2}{\gamma_{\min}} \ln \frac{1}{\delta}, \end{aligned} \quad (26)$$

the last term of which is $H^2/\gamma_{\min} = \tilde{\mathcal{O}}(1/\eta_K)$ (the extra factor of H from the magnitude bound is incorporated into the $\tilde{\mathcal{O}}$ notation). The factor of H in front of the variance, paired with $\lambda = \gamma_{\min}/H^2$, leaves the variance contribution of order $\gamma_{\min}H/(\bar{q}_k + \gamma_k)$, identical to the per-step Freedman bound used in the original analysis.

We now turn to the expected-bias piece. By virtue of the law of total expectation in tandem with $\mathbb{E}_{k-1}[L_{k,h} \mathbf{1}\{B = B_h^k\}] = q_k(B) Q_{k,h}^{\pi_k}(B)$,

$$\mathbb{E}_{k-1} \left[\sum_B q_{k,h}^{\star, \pi_k}(B) \widehat{Q}_{k,h}(B) \right] = \sum_B q_{k,h}^{\star, \pi_k}(B) \frac{q_k(B)}{\bar{q}_k(B) + \gamma_k} Q_{k,h}^{\pi_k}(B),$$

whence

$$\begin{aligned} & \sum_B q_{k,h}^{\star, \pi_k}(B) Q_{k,h}^{\pi_k}(B) - \mathbb{E}_{k-1} \left[\sum_B q_{k,h}^{\star, \pi_k}(B) \widehat{Q}_{k,h}(B) \right] \\ & = \sum_B q_{k,h}^{\star, \pi_k}(B) Q_{k,h}^{\pi_k}(B) \cdot \frac{\bar{q}_k(B) + \gamma_k - q_k(B)}{\bar{q}_k(B) + \gamma_k} \\ & \leq \sum_B q_{k,h}^{\star, \pi_k}(B) \frac{H(\bar{q}_k(B) - \underline{q}_k(B) + \gamma_k)}{\bar{q}_k(B) + \gamma_k}, \end{aligned}$$

where the inequality follows from $Q_{k,h}^{\pi_k}(B) \leq H$ in tandem with $\underline{q}_k(B) \leq q_k(B) \leq \bar{q}_k(B)$, the latter holding on the high-probability confidence event $\mathcal{P} \in \mathcal{P}_j$.

Combining Eq. (26) with the expected-bias bound, the $\gamma_{\min}H/(\bar{q}_k + \gamma_k)$ contribution from the martingale and the $H\gamma_k/(\bar{q}_k + \gamma_k)$ contribution from the expected bias merge into $H(2\gamma_k + \bar{q}_k - \underline{q}_k)/(\bar{q}_k + \gamma_k)$, using $\gamma_{\min} \leq \gamma_k$ and noting that $Q_{k,h}^{\pi_k} \leq H$ was already incorporated into the $H\gamma_k$ numerator, so there is no additional factor of H . Hence

$$\text{BIAS-1} \leq \tilde{\mathcal{O}} \left(\frac{1}{\eta_K} \right) + \sum_{k=1}^K \sum_{h=1}^H \sum_{B \in P_h^k} q_{k,h}^{\star, \pi_k}(B) \frac{2\gamma_k H + H(\bar{q}_k(B) - \underline{q}_k(B))}{\bar{q}_k(B) + \gamma_k},$$

as desired. $\square$

Lemma 12. *With probability at least $1 - 5\delta$, we obtain*

$$\text{BIAS-2} \leq \tilde{\mathcal{O}}\left(\frac{1}{\eta_K}\right) + \sum_{k,h,B} q_{k,h}^{\star,\pi^*}(B) b_{k,h}(B).$$

Proof. Adding and subtracting the conditional expectation given the history prior to episode k (denoted by $\mathbb{E}_{k-1}[\cdot]$) produces the per-episode decomposition

$$\begin{aligned} \text{BIAS-2} &= \sum_{k=1}^K \sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) (\hat{Q}_{k,h}(B) - Q_{k,h}^{\pi_k}(B)) \\ &= \underbrace{\sum_{k=1}^K \left(\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) \hat{Q}_{k,h}(B) - \mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) \hat{Q}_{k,h}(B) \right] \right)}_{\text{deviation}} \\ &\quad + \underbrace{\sum_{k=1}^K \left(\mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) \hat{Q}_{k,h}(B) \right] - \sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) Q_{k,h}^{\pi_k}(B) \right)}_{\text{expected bias}}. \end{aligned} \quad (27)$$

As in Lemma 11, the per-episode aggregates form a martingale difference sequence with respect to the natural filtration, whereas the per-layer aggregates do not. We first show that the expected-bias piece is nonpositive on the high-probability confidence event, then bound the deviation piece by Freedman's inequality.

For the expected-bias piece, the law of total expectation in tandem with $\mathbb{E}_{k-1}[L_{k,h} \mathbf{1}\{B = B_h^k\}] = q_k(B) Q_{k,h}^{\pi_k}(B)$ delivers, for each layer h ,

$$\mathbb{E}_{k-1}[\hat{Q}_{k,h}(B)] = \frac{q_k(B)}{\bar{q}_k(B) + \gamma_k} Q_{k,h}^{\pi_k}(B).$$

On the confidence event $\mathcal{P} \in \mathcal{P}_j$ for all j , which holds with probability at least $1 - \delta$ by virtue of Lemma 5 and a union bound over all triples (h, B, B') , the upper occupancy satisfies $\bar{q}_k(B) \geq q_k(B)$, whence the ratio $q_k(B)/(\bar{q}_k(B) + \gamma_k) \leq 1$, and summing over h ,

$$\begin{aligned} &\mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) \hat{Q}_{k,h}(B) \right] - \sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) Q_{k,h}^{\pi_k}(B) \\ &= \sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^*}(B) Q_{k,h}^{\pi_k}(B) \left(\frac{q_k(B)}{\bar{q}_k(B) + \gamma_k} - 1 \right) \\ &\leq 0. \end{aligned}$$

For the deviation piece, the magnitude bound on the per-episode aggregate follows from the indicator of $\widehat{Q}_{k,h}$: each layer h contributes at most one nonzero summand, whence

$$0 \leq \sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^\star}(B) \widehat{Q}_{k,h}(B) \leq H \cdot \frac{H}{\gamma_{\min}} = \frac{H^2}{\gamma_{\min}},$$

by virtue of $q_{k,h}^{\star,\pi^\star}(B_h^k) \leq 1$, $L_{k,h} \leq H$, and $\bar{q}_k(B_h^k) + \gamma_k \geq \gamma_{\min}$. The conditional second moment satisfies

$$\begin{aligned} & \mathbb{E}_{k-1} \left[\left(\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^\star}(B) \widehat{Q}_{k,h}(B) - \mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^\star}(B) \widehat{Q}_{k,h}(B) \right] \right)^2 \right] \\ & \leq H \sum_{h=1}^H \mathbb{E}_{k-1} \left[\left(\sum_B q_{k,h}^{\star,\pi^\star}(B) \widehat{Q}_{k,h}(B) \right)^2 \right] \\ & \leq H \sum_{h=1}^H \sum_B (q_{k,h}^{\star,\pi^\star}(B))^2 \frac{H^2 q_k(B)}{(\bar{q}_k(B) + \gamma_k)^2} \\ & \leq H^3 \sum_{h=1}^H \sum_B \frac{q_{k,h}^{\star,\pi^\star}(B)}{\bar{q}_k(B) + \gamma_k}, \end{aligned}$$

where the second-to-last inequality relies on $L_{k,h} \leq H$ in tandem with $q_{k,h}^{\star,\pi^\star}(B) \leq 1$, and the last on $q_k(B) \leq \bar{q}_k(B) + \gamma_k$.

Invoking Lemma 4 on this per-episode martingale difference sequence with $b = H^2/\gamma_{\min}$ and $\lambda = \gamma_{\min}/H^2$ (verifying the precondition $\lambda b = 1 \leq 1$) produces, with probability at least $1 - \delta$,

$$\begin{aligned} & \sum_{k=1}^K \left(\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^\star}(B) \widehat{Q}_{k,h}(B) - \mathbb{E}_{k-1} \left[\sum_{h=1}^H \sum_B q_{k,h}^{\star,\pi^\star}(B) \widehat{Q}_{k,h}(B) \right] \right) \\ & \leq \frac{\gamma_{\min}}{H^2} \cdot H^3 \sum_{k,h,B} \frac{q_{k,h}^{\star,\pi^\star}(B)}{\bar{q}_k(B) + \gamma_k} + \frac{H^2}{\gamma_{\min}} \ln \frac{1}{\delta} \\ & = \gamma_{\min} H \sum_{k,h,B} \frac{q_{k,h}^{\star,\pi^\star}(B)}{\bar{q}_k(B) + \gamma_k} + \frac{H^2}{\gamma_{\min}} \ln \frac{1}{\delta}. \end{aligned}$$

The first contribution merges into the bonus on the right-hand side of Eq. (28): indeed, $\gamma_{\min} \leq \gamma_k$ in tandem with the inequality $\gamma_k H / (\bar{q}_k(B) + \gamma_k) \leq b_{k,h}(B)$ from Eq. (2) ensures

that

$$\gamma_{\min} H \sum_{k,h,B} \frac{q_{k,h}^{*,\pi^*}(B)}{\bar{q}_k(B) + \gamma_k} \leq \sum_{k,h,B} q_{k,h}^{*,\pi^*}(B) b_{k,h}(B),$$

which is subsumed when Proposition 13 consolidates BIAS-1, BIAS-2, and REG-TERM. The second contribution is $H^2/\gamma_{\min} = \tilde{\mathcal{O}}(1/\eta_K)$.

Combining the deviation piece with the (nonpositive) expected bias on the high-probability event,

$$\text{BIAS-2} \leq \sum_{k,h,B} q_{k,h}^{*,\pi^*}(B) b_{k,h}(B) + \tilde{\mathcal{O}}\left(\frac{1}{\eta_K}\right),$$

as desired. □

A.8.3 Combined Three-Term Decomposition

Proposition 13 (Ball-level analogue of Eq. (5) in Luo et al. 2021). *Under the confidence event, with probability at least $1 - 5\delta$, we obtain*

$$\begin{aligned} & \sum_{k=1}^K \sum_{h=1}^H \sum_{B \in P_h^k} \left(q_{k,h}^{*,\pi_k}(B) - q_{k,h}^{*,\pi^*}(B) \right) \left(Q_{k,h}^{\pi_k}(B) - \mathcal{B}_{k,h}(B) \right) \\ & \leq \tilde{\mathcal{O}}(H/\eta_K) + \sum_{k=1}^K \sum_{h=1}^H \sum_B q_{k,h}^{*,\pi_k}(B) b_{k,h}(B) \\ & \quad + \sum_{k=1}^K \sum_{h=1}^H \sum_B q_{k,h}^{*,\pi^*}(B) b_{k,h}(B) + \frac{1}{H} \sum_{k=1}^K \sum_{h=1}^H \sum_B q_{k,h}^{*,\pi_k}(B) \mathcal{B}_{k,h}(B). \end{aligned} \quad (28)$$

Proof. Inserting $\pm \hat{Q}_{k,h}(B)$ inside the parentheses on the left-hand side of Eq. (28) decomposes it into the sum BIAS-1 + BIAS-2 + REG-TERM of Eq. (7). Invoking the three previously established bounds,

$$\begin{aligned} \text{BIAS-1} & \leq \tilde{\mathcal{O}}\left(\frac{1}{\eta_K}\right) + \sum_{k,h,B} q_{k,h}^{*,\pi_k}(B) b_{k,h}(B), \\ \text{BIAS-2} & \leq \tilde{\mathcal{O}}\left(\frac{1}{\eta_K}\right) + \sum_{k,h,B} q_{k,h}^{*,\pi^*}(B) b_{k,h}(B), \\ \text{REG-TERM} & \leq \tilde{\mathcal{O}}\left(\frac{H}{\eta_K}\right) + \sum_{k,h,B} q_{k,h}^{*,\pi_k}(B) \left(\frac{\gamma_k H}{\bar{q}_k(B) + \gamma_k} + \frac{\mathcal{B}_{k,h}(B)}{H} \right), \end{aligned}$$

where the BIAS-1 bound matches the form of $b_{k,h}(B)$ in Eq. (2) term-by-term, the H factor from $Q^{\pi_k} \leq H$ being already incorporated inside $b_{k,h}$ rather than appearing as an additional multiplier. The one-step bonus $b_{k,h}(B)$ incorporates the $\gamma_k H / (\bar{q}_k(B) + \gamma_k)$ contribution from REG-TERM by construction, since $b_{k,h}(B) \geq \gamma_k H / (\bar{q}_k(B) + \gamma_k)$ for every ball. The BIAS-2 bonus term, which carries q^{*,π^*} weighting from the Freedman variance bound, appears on the right-hand side of Eq. (28) alongside the q^{*,π_k} -weighted terms. Grouping the $\tilde{\mathcal{O}}(H/\eta_K)$ contributions, the REG-TERM term being much larger, we obtain Eq. (28) as desired. $\square$